

CHAPTER

1

SCOPE AND ADMINISTRATION

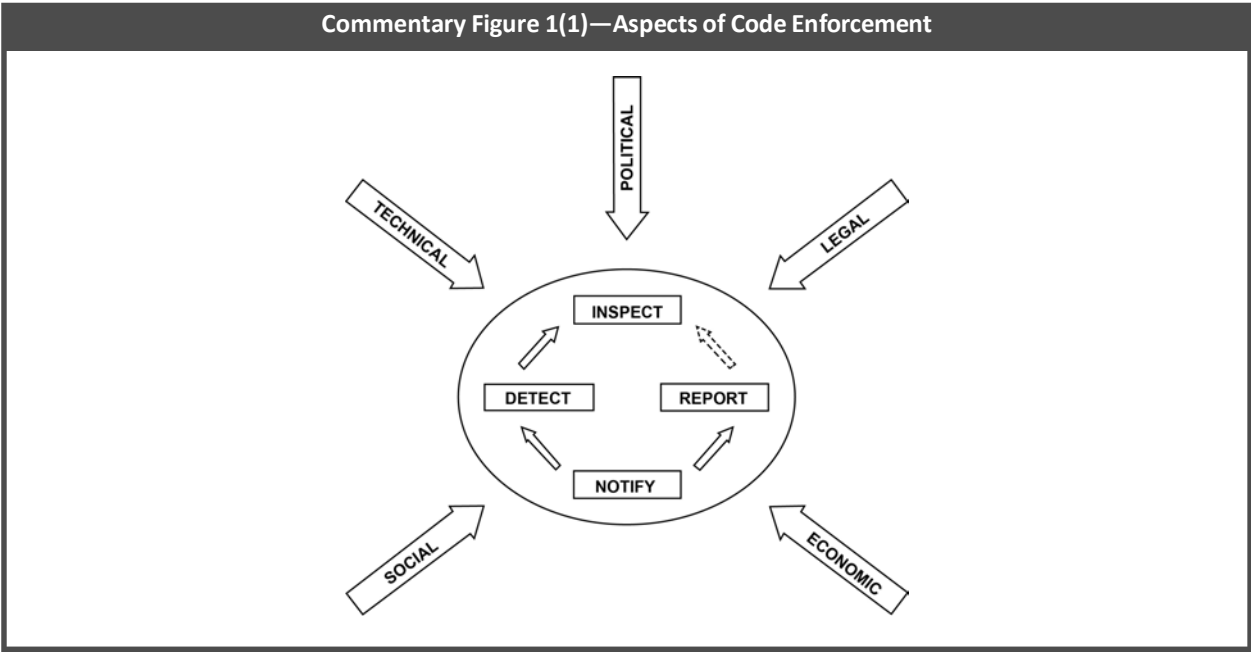
General Comments

This chapter addresses the administration and enforcement of the code. The objectives and mandate for enforcement are beyond the scope of this chapter. Before adopting the code, a state or local government must establish an agency with staff trained to administer and enforce the code. The administrative relationships, designation of the enforcement authority (fire code official), funding, training and certification of inspectors and scope of the enforcement program are determined by the adopting body.

Management personnel generally perform functions, such as planning, organizing, directing, controlling, analyzing and budgeting. Though the code administrator’s duties may include all of these functions, this chapter takes a much narrower view of the code administrative function, dealing mainly with technical and legal areas. Fire prevention code administration must be considered in the context of a complex environment containing political, social, economic, technical and legal aspects. Enforcement, too, is a broad term that includes a range of activities aimed at identifying and eliminating hazards; in this case, hazards causing or contributing to a fire or impairing life safety.

Four functions are commonly associated with enforcement: inspecting, detecting, notifying and reporting [see Commentary Figure 1(1)]. This chapter describes the technical and legal requirements associated with administering a code enforcement program to achieve these functions. The examination of these concepts specifically provides a better understanding of the fire code official’s authority, duties and liabilities.

Two main duties of the fire code official are administration and enforcement. In administration, the following concepts are most important:



Code Administrative Environment

Many administrative or management functions are not addressed in the code. Before the provisions of this document can be of any use, many basic questions must be answered. Jurisdictions adopting a code enforcement program are using discretionary powers to fulfill a community need. The need in the community must be clearly identified, the program mission clearly established and the most appropriate delivery system selected. To address the technical and legal demands of the code administrative environment, the code assumes that jurisdictions adopting the document are interested in protecting the health, safety and welfare of its citizens from the effects of fires and explosions. Additionally, the code assumes that these jurisdictions are authorized to use the police power of the state to receive these benefits. Finally, the code assigns principal responsibility for enforcing this document to the department or agency (fire department or fire prevention bureau) most frequently available to perform this mission.

The particular objectives and social or political mandate of a code enforcement program are not considered in the context of this document. These items, however, are often cited as the most frustrating problems faced by code administrators. Code enforcers often complain of being overwhelmed by demands for leniency or special consideration based on the economic, social or political effects of their decisions.

As stated, this chapter establishes ground rules for enforcing the code; however, these ground rules are only the technical and legal requirements binding both fire code officials and the general public. For guidance on the political, social and economic considerations associated with code enforcement activities, adopting authorities must turn elsewhere; however, none of this is intended to imply that these considerations are absent from the code process. To the contrary, by establishing these requirements as “minimums,” the ICC voting membership (see ICC Board of Directors Policy CP-28) has, through a democratic process of public hearings and debate, attempted to weigh these considerations carefully when deliberating, modifying and adopting the provisions appearing in this document. In the end, each jurisdiction must give careful consideration to how these requirements should be adopted; who should be responsible for enforcing them; how enforcement personnel should be trained; how the operation will be financed; and when and how to modify or change operations, if necessary. These considerations deserve careful, thorough public attention before a decision is made to adopt and enforce the code.

Scope and Applicability of the Code

The code applies to new and existing structures and premises as prescribed in Sections 102.1 and 102.2, in matters related to occupancy and maintenance for the protection of lives and property from fire. Conditions related to fire spread, occupant hazard protection and maintenance are regulated as follows:

Retroactivity: Because the code applies to both new and existing structures and premises as prescribed in Sections 102.1 and 102.2, the existing building provisions may be considered retroactive. Existing structures and premises built in compliance with the codes and standards in effect at the time of their original construction or alteration are not in all cases exempt from code compliance.

Other codes and standards: The code relies heavily on other codes and standards to specify a means of complying with its provisions, including, among others, the *International Building Code*® (IBC®), the *International Mechanical Code*® (IMC®), the *International Fuel Gas Code*® (IFGC®) and the standards referenced in the text. Additionally, other federal, state and local codes and ordinances may establish certain requirements related to fire protection and life safety. Code requirements are intended to complement other regulations. Where conflicts arise between code provisions and the referenced standards, the code provisions apply. Where a standard provides additional technical detail or guidance beyond that provided in the related code text, the fire code official must use judgment when applying these provisions to prevent conflicts with the code provisions. If a conflict arises, it is the fire code official's duty to determine which provisions secure the code's intent. When a conflict between codes or other legal action causes a portion of this document to be “struck down,” such action is not intended to invalidate the remaining code provisions. The severability of code provisions, however, does not imply that these same provisions should be considered or applied outside of their context as a part of the code.

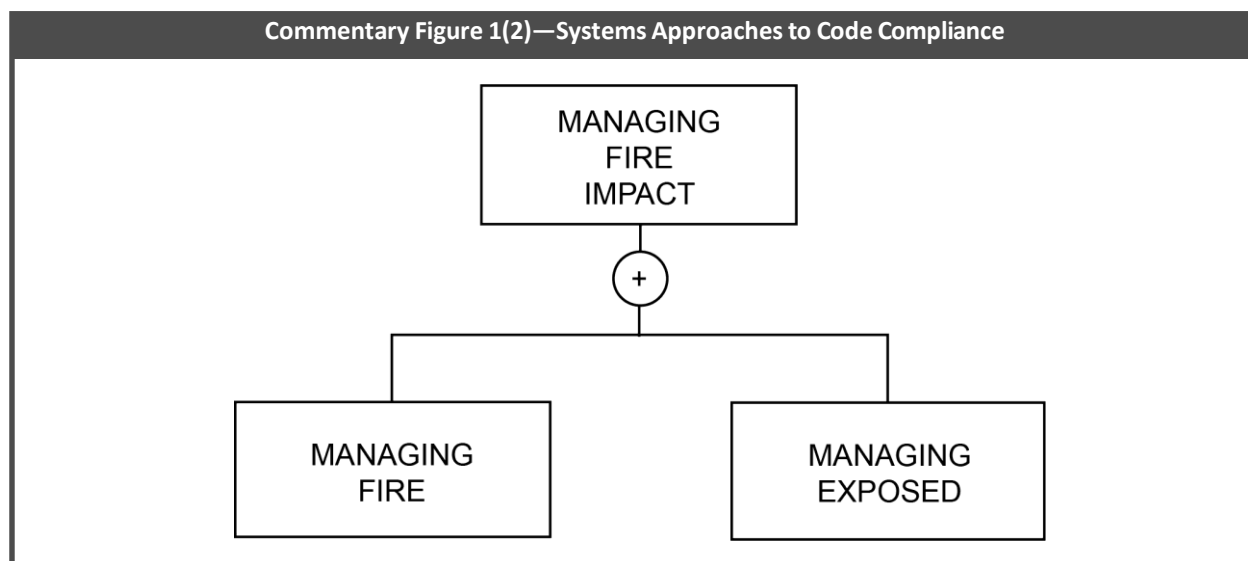
Fire Code Official's Judgment

The code relies heavily not only on other codes and standards but also on the judgment and experience of the fire code official.

Approval: The code details occupancy and maintenance requirements; however, it relies heavily on performance criteria, as opposed to detailed specifications, to accomplish this task. The fire code official, therefore, must exercise judgment when approving or permitting operations, processes and procedures required by the code. Proof of compliance may include certification or labeling by independent testing laboratories; however, regardless of the conclusions of these external agencies and authorities, the fire code official remains the sole judge of what fulfills the intent of the code. This becomes particularly important when the fire code official is asked to evaluate equivalent methods and materials. Having piles of data may seem helpful, but the data may prove to be meaningless when it is considered in the context of the code's intent. Data in support of alternative methods and materials must demonstrate not only compliance with the code's intent but also relevance to the issues at hand. Evidence, such as a label or an independent laboratory test report, may sometimes be used inappropriately to support an application for recognition of equivalency. The fire code official must evaluate all submitted evidence to make sure it applies to its intended use, as well as to the code's intent. In an increasingly technical and litigious society, learning how to make such judgments may be the biggest challenge facing fire code officials. Relying on strict interpretations of intent or the “letter” of the code may be the conservative way, but conservative approaches may simply increase the social and political pressures confronting fire code officials. Decision-making aids employing contemporary computer technology permit a designer to propose more innovative and creative responses to complex problems. Fire code officials must begin to recognize, use and interpret these tools and data to maintain effective protection. These models permit designers to quickly and easily evaluate the relationships and performance of a variety of complex variables.

Another model that does not rely on a computer is NFPA 550 [see Commentary Figure 1(2)]. This model requires little training to use or understand and is an all-inclusive representation of the variables contributing to fire safety. The model

may, therefore, serve as a useful tool for qualitatively evaluating the contribution of various approaches to an overall fire safety system. Once equivalent alternative methods have been identified using the Fire Safety Concepts Tree (see “General Comments” in Chapter 3), quantitative (cost/benefit) analyses may be applied.



Fire Code Officials and Liability

Like all professionals, fire code officials are subject to legal action. The two most common legal actions that may be pursued against fire code officials are breach of contract lawsuits and tort claims. Tort claims, by far, are the most common lawsuits. These lawsuits allege that some damage, injury or harm (a tort) resulted from the actions of the fire code official. A successful tort claim must prove that the plaintiff was injured or harmed; that the fire code official had a legal duty or obligation to perform with respect to the plaintiff and that the cause of the plaintiff's injury was the fire code official's actions or inactions while performing these duties.

The Law of Torts includes the following:

The tort: Damages arising from the acts of fire code officials fall into two broad categories: property and personal [see Commentary Figure 1(3)]. Property torts involve the control, use, operation or ownership of personal and real property by private individuals. Personal torts involve physical, verbal or written assaults on the character, person, psyche or privacy of individuals. Such assaults or invasions may involve actual contact or threat of harm. For example, fire code officials' acts of commission may restrain business or trade activity, while acts of omission may fail to recognize that hazards need to be corrected, thus resulting in life or property losses.

Two actions dominate lawsuits filed against enforcement authorities: Most lawsuits either allege improper acts by the fire code official (acts of commission) or failure to fulfill specified or implied legal obligations (acts of omission). In the former, plaintiffs usually seek temporary or permanent relief from a fire code official's decision. In these actions, plaintiffs usually allege improper interpretation or application of the code or its intent. Other lawsuits usually allege failure to exercise a reasonable standard of care in the performance of duties of the fire code official. In either type of lawsuit, and often in the case of omissions, plaintiffs seek compensatory and even punitive damages. Infringements on constitutional protections may be, though occurring infrequently, the basis for lawsuits against fire code officials. Common constitutional issues raised in lawsuits against fire code officials include violations of the Fourth Amendment's protection against unreasonable searches and seizures, the Sixth Amendment's due process protections and the Fourteenth Amendment's equal protection provisions. First Amendment rights guaranteed under freedom of association protections may be raised in cases involving public assembly occupancies, especially churches.

Condition of negligence: To prevail in a tort claim action, a plaintiff must demonstrate negligence on the part of the defendant. Negligence may be simple—a failure to exercise reasonable or adequate care when performing assigned duties (commonly known as misfeasance)—or it may be gross—represented by wanton, willful, reckless or malicious disregard for public safety. Criminal activities, including dereliction (nonfeasance) or the failure to perform required assigned duties, may be cause for claims of gross negligence. Likewise, malfeasance, the willful or malicious violation of a legal duty, may constitute grossly negligent behavior. The following three elements must be proven to sustain a claim of negligence: the defendant had a duty to act, the defendant failed to exercise the required standard of care in the performance of that duty and, as a result of that failure, damage was incurred or harm was suffered by the plaintiff.

Duty to act: The code establishes few duties of the fire code official; instead, it places greatest emphasis on the responsibility of structure or premises owners and operators to perform their duties with adequate regard for public health, safety and welfare. The duties owed to the public by the fire code official fall under the following categories: approvals, enforcement, personnel, inspections, investigations, reports and record keeping. Other duties may be assumed by fire code officials through the performance of their official duties. Some courts have ruled that failure to perform timely reinspections or exhaust legal remedies against violators in fire code cases creates a special relationship between the fire code official and the occupants of properties in violation of the code, especially when the occupants do not own the property and are not responsible for code compliance. Some court rulings have even implied that conducting inspections not otherwise required by the code constitutes an *ultra vires* (beyond the authority of) liability. Fire code officials should consult their jurisdiction's legal counsel to determine how these decisions, the jurisdiction's enforcement policies and the code provisions combined affect their enforcement program and jurisdictional and personal liabilities.

Standard of care: Taken together, the fire code official's duties are the basis for determining their standard of care. When assessing whether fire code officials have met this standard, judges and juries must determine whether they performed the required duties as reasonable, comparably trained and experienced fire code officials. Failure to meet the appropriate standard of care may be classified in three ways: nonfeasance, misfeasance or malfeasance. Nonfeasance is the failure to perform a required duty. Improper performance of a required duty constitutes misfeasance, and malicious or willful violation of a required duty is malfeasance. Of the three, misfeasance or simple negligence is the most common cause of action. The code and most tort claims either hold the government immune from specific claims of misfeasance or severely limit damage awards in such cases. For all purposes, sovereign immunity—the doctrine inherited from British common law mandating that “the King can do no wrong”—is obsolete. Similarly, courts in many states have abandoned the public duty doctrine, which states that a duty to all is a duty to no one. Holding that most code provisions and governmental regulations secure benefits for select groups, some state courts recognize that specific enforcement activities secure greater benefits for some members of the public than others. Such judicial reasoning holds that the inspector's duty applies to the individual who may be injured as a result of failure to detect a hazard or diligently pursue compliance. Moreover, this duty may include acts of omission, such as failure to perform required inspections. With courts today recognizing only limited immunity for government officials, fire code officials must become more aware of their duties and liabilities. Although tort claim acts limit damage awards, they still permit lawsuits to proceed against governmental officials and agencies to determine their responsibility for negligent acts. Claims of gross negligence arising from nonfeasance or malfeasance are less common than misfeasance actions but are predictably harder to defend. The code provides no relief from liability where the fire code official either fails to perform a required duty or acts *ultra vires*; that is, beyond his or her authority. The jurisdiction is generally immune from claims when its agents perform acts beyond the scope of their authority, unless such acts were implicitly endorsed by the government (explicit endorsement may constitute a discretionary governmental act and, similarly, immunize the government). Nonfeasance is considered a criminal offense in many jurisdictions. An employee's dereliction of duty exempts the jurisdiction from immunity under most circumstances, unless the employee's failure to perform was the direct result of explicit instructions from governmental superiors; however, the employee may be held criminally liable.

In addition to the Law of Torts, the following have an impact on fire code officials and liability:

Awards: Lawsuits may seek declarative judgments (injunctive relief or monetary awards) in favor of the plaintiff. Monetary awards fall into four categories: nominal, special, compensatory and punitive. The first purpose of monetary awards should be to the claimant or plaintiff for real losses. This is the purpose of compensatory and special damages. Compensatory awards reimburse the claimant or plaintiff for the direct costs resulting from the defendant's negligence or carelessness. Many times, a plaintiff will also seek additional compensation for the indirect results of the defendant's acts. Such special damage claims may result in additional compensation beyond that provided by compensatory damages. Punitive awards are intended to punish the defendant for the misdeed and discourage them from future unlawful activity. These awards are often held up as examples to the community as a whole and are a way to discourage unlawful activities by others. Nominal damage awards serve to assign blame in intentional tort cases when the facts of the case do not merit a substantial settlement.

Protection: The best protection against a lawsuit is professional conduct and preparation; that is, training, education and research. Lawsuits filed against public officials have become commonplace and are probably inevitable. In 1983, H. M. Markman suggested six rules to manage legal liability [see Commentary Figure 1(4)].

Although no single rule should be considered more important than another, the last one is perhaps the best to remember. Everyone makes mistakes, so strive to learn from the mistakes rather than repeat them. Nonetheless, every mistake may be potential exposure. Acting professionally helps minimize exposure to error, especially when training, and common sense is encouraged. Using common sense, exercising reasonable care and acting professionally are no insurance against a lawsuit, but they all may provide considerable protection in the event a lawsuit is filed. No matter how hard someone may try to avoid a lawsuit, someone may sue. When a lawsuit is filed, the most important things to remember are not to take it personally and not to forget the other five rules.

Commentary Figure 1(3)—Types of Torts

PROPERTY	PERSONAL
Trespass Conversion Nuisance	Assault and Battery False Arrest or Imprisonment Defamation, Slander and Libel

Source: Rosenbauer, D.L., *Introduction to Fire Protection Law*.

Commentary Figure 1(4)—Markman's Six Rules

- You cannot prevent someone from filing a lawsuit against you.
- Do not take the lawsuit personally.
- Understand your risk exposure or exposures.
- Be professional.
- You are not an insurer.
- Do not make stupid mistakes.

Enforcement

The enforcement duty of the fire code official's position is composed of four distinct functions: inspection, detection, notification and reporting. All four functions define phases in the enforcement process duties of fire code officials.

During the code enforcement process, structures or premises requiring inspections are identified. Inspectors are assigned and inspections are performed. During these inspections, any code violations found are usually noted. Then, the owner or occupant is verbally advised or notified that the deficiencies noted are code violations. To promote code compliance, the inspector may suggest remedial actions that may be taken to establish compliance. Finally, a written violation notice serving as further notice to the owner or occupant is issued. The written notice also serves as a permanent record or report of the inspection.

Inspection

Inspections are careful examinations of plans or premises for the presence of fire and life safety hazards. Upon observing a hazardous condition, the fire code official begins a process directed at correcting the situation. This may be accomplished by removing or eliminating the hazardous condition or providing some countermeasure designed to lessen its effects on the property, occupants or neighbors. Inspections should demonstrate a systematic method that keeps the inspection process in a proper perspective and recognizes that code enforcement is limited to legal and technical means of pursuing fire safety. Achieving fire safety objectives means using a balanced approach composed of some elements seeking to prevent ignitions and others attempting to control fire effects. Fire safety objectives are not defined by the code but rather by the users. Each jurisdiction must establish what risks and costs are reasonable while pursuing fire safety.

There may be as many different methods of conducting inspections as there are inspectors. No single method is necessarily the correct one; however, each method probably has some strong and weak points. The following three approaches can form the basis for any number of different inspection techniques.

Outside to inside: Beginning outside is not only logical but necessary. Inspectors too often neglect hazards and clues outside the building that suggest significant danger to the occupants. An inspector must ask the following question: "Do the things I see outside match those I see inside?" For example:

- Do doors identified as exits inside actually discharge outside to acceptable refuge areas or the public way?
- Are trash receptacles or other obstructions outside located so the effectiveness of exits is not reduced, or do receptacles alone pose a fire hazard?
- Does the site permit sufficient access for firefighter rescue operations and fire suppression?

Top to bottom: Once inside a structure, deciding where to start is more than a matter of preference. One question inspectors may ask is: "If completeness is the principal criterion, why not start at the bottom and work up?" The answer to this question is that walking down stairs is easier than walking up stairs. This easier path of travel allows the inspector to concentrate more completely on the inspection itself. After performing many inspections, there will not be a need for additional exercise obtained from beginning at the bottom.

General to specific: Without constructing a detailed inspection framework, many fire code officials find it helpful to move from the general to the specific when evaluating occupancies and hazards. This helps keep the whole problem in focus while preserving attention to detail. The inspector can focus on a specific problem without losing sight of the "big picture."

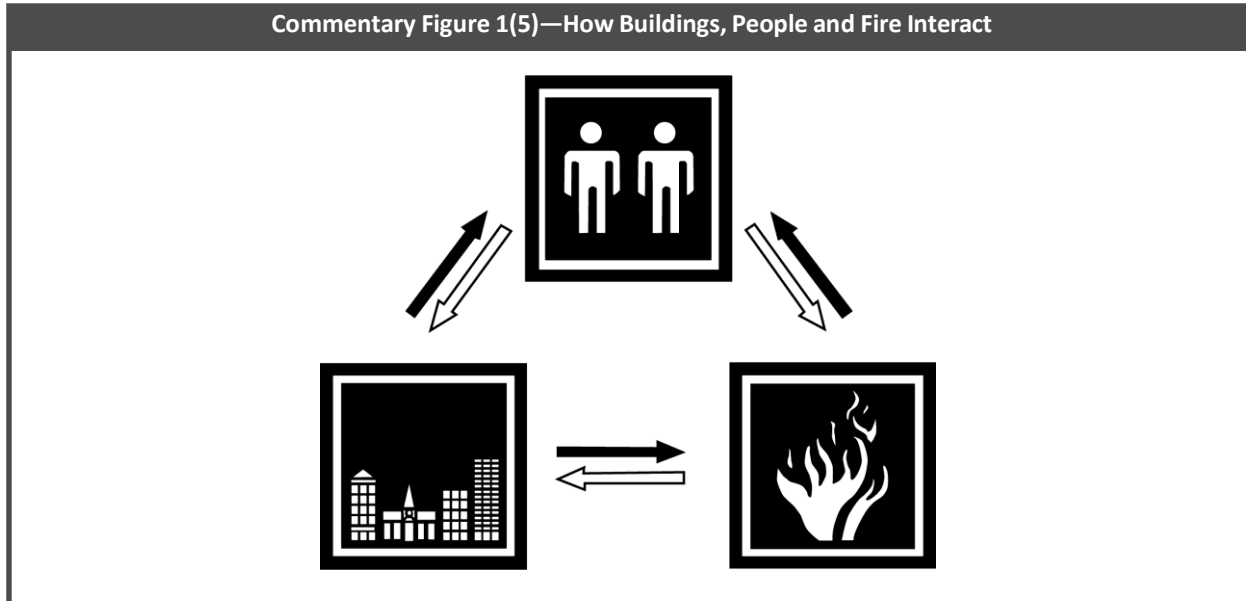
Detection

Systematic inspection procedures, like those described, should aid in the detection of code violations. By keeping the premises, processes or objectives in clear focus, the inspector keeps the task in context. A systematic inspection process

implies not only organization but an understanding of the entire process. Achieving fire safety objectives means understanding how structures, premises, occupants and fire interact [see Commentary Figure 1(5)]. More specifically, each aspect affects another. Human behavior affects the likelihood of fire and how humans react to fire. The type of facility will have an impact on the types of fires and types and characteristics of occupants. To keep the system in balance and prevent uncontrolled fires means understanding how people use structures and premises.

In such a context, a fire hazard is anything that either fails to prevent an uncontrolled fire or permits a fire to spread unchecked. Similarly, hazardous conditions are those preventing occupants from escaping or firefighters from entering a structure and premises to control a fire.

Commentary Figure 1(5)—How Buildings, People and Fire Interact



Notification

Inspection programs cannot identify and abate all hazards. Code enforcement alone cannot secure absolute protection for people and property. Further, many code requirements maintain or reinforce features not intended to prevent a fire but rather to minimize a fire's effects should one occur. Every inspection program, therefore, should consider the benefits of educating building owners and occupants about the hazards endangering their lives and property. Not only do such efforts help secure compliance with code requirements, but they are likely to secure long-term commitments to fire safety as well. Another equally apt metaphor describes the fire prevention process as the "Three Es": engineering, education and enforcement. A balanced approach composed of these three elements can be an especially effective way of achieving desired fire safety objectives.

Reporting

The first three elements of the code enforcement process are directed at identifying and eliminating hazards at their source. Reporting is intended to help document and reinforce the lessons learned from the previous three phases. The words, "If it's not written down, it didn't happen!" reinforce the message that reporting is just as important as any of the other three elements of the code enforcement system. Few people enjoy paperwork and data entry. Without documentation, however, prosecuting an effective code enforcement program becomes nearly impossible. Accurate, concise and timely records are essential for both legal and historical reasons. Documenting the inspection and violation history of a particular premises or owner is essential when prosecuting criminal actions under the code provisions.

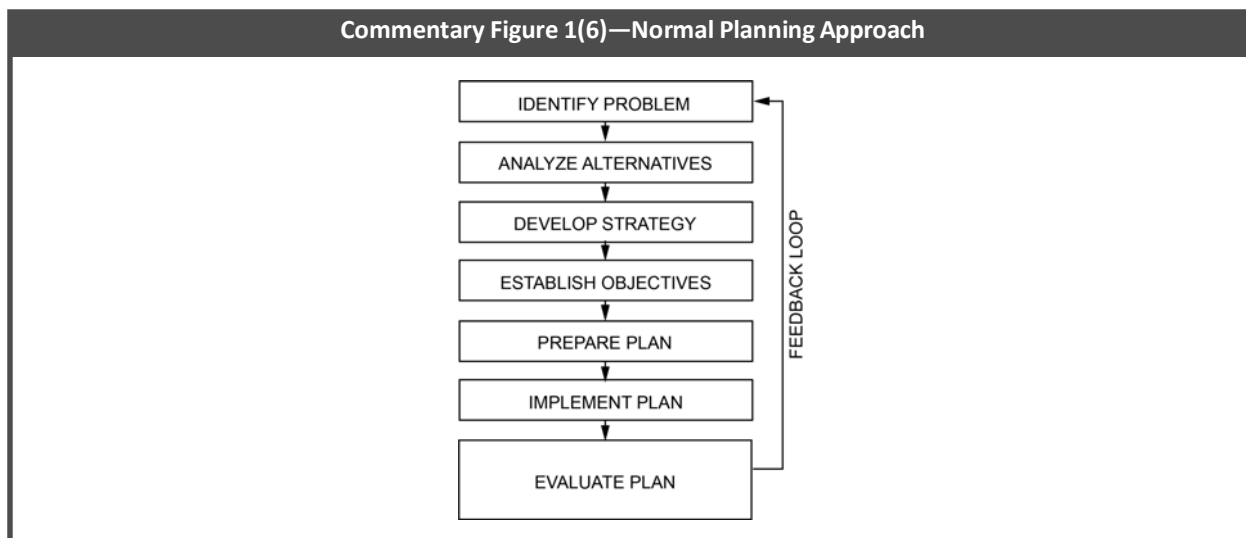
Commentary Figure 1(6) illustrates a systems approach using data generated by fire incidents and inspections to direct code enforcement, public education activities or code development. This approach is equivalent to the one typically used to make ordinary decisions about problems with many competing solutions, to plan for the future and to consider the cost and benefits of these decisions.

This system is used mostly to solve everyday problems, weigh costs and benefits of alternatives, and plan responses to problems. It is also a helpful tool for responding to fire protection problems. As an example, fire data suggest the emergence of a trend toward more fires caused by misuse of auxiliary heating appliances. Alternative solutions include:

- A public education campaign on safe use of heaters.
- A targeted inspection program to identify and correct improper heating practices in homes and businesses.
- Doing nothing but hoping for warm weather.

The evaluation must consider a variety of complex factors and methods of measuring the success of each approach. Once an approach is selected and implemented, it must be evaluated to examine whether the desired reduction in fires occurs. If not, perhaps one of the other alternatives should be reexamined, or the problem should be redefined.

Understanding the code administration process and the environment influencing it allows the fire code official to be more effective. Adhering to the provisions of Chapter 1 not only minimizes the fire code official's liability, but also provides an effective code enforcement program. Just as owners and occupants have obligations under the code, so does the fire code official. Following these procedures enables them to identify and respond to the community's needs, thus reducing the community's fire risk.



Purpose

Chapter 1 establishes provisions to ensure that code administration and enforcement is reasonable, appropriate and fair. This chapter outlines the duties and powers of the fire code official; the scope of the fire code official's authority to enforce the code; the applicability of the document and proofs of compliance; the means of securing compliance with its provisions and procedures for protecting due process rights of applicants, owners, occupants and others affected by the code provisions; and the enforcement activities of the fire code official.

PART 1—GENERAL PROVISIONS

SECTION 101—SCOPE AND GENERAL REQUIREMENTS

[A] 101.1 Title. These regulations shall be known as the *Fire Code* of [NAME OF JURISDICTION], hereinafter referred to as “this code.”

[C] This section identifies jurisdictional applicability in legal terms. The local jurisdiction is to insert its name into this section by including a modification to the code in the adopting ordinance. This will make the code applicable to the local jurisdiction.

[A] 101.2 Scope. This code establishes regulations affecting or relating to structures, processes, premises and safeguards regarding all of the following:

1. The hazard of fire and explosion arising from the storage, handling or use of structures, materials or devices.
2. Conditions hazardous to life, property or public welfare in the occupancy of structures or premises.
3. Fire hazards in the structure or on the premises from occupancy or operation.
4. Matters related to the construction, extension, repair, *alteration* or removal of fire protection systems.
5. Conditions affecting the safety of firefighters and emergency responders during emergency operations.

[C] The code does not attempt to achieve perfection by requiring every conceivable or available safeguard for every structure, premises or operation within the scope of the code; rather, the code seeks to establish a minimum acceptable safety level to balance the many factors that must be considered, including loss statistics, relative hazard and the economic and social impact. The code is maintained through the use of a democratic code development process so that everyone affected by these minimum requirements has an equal opportunity to present his or her concern, both for and against any of the requirements.

The question is often asked, “Does the code apply only to buildings and facilities, or does it cover vehicles as well?” Though the scope text does not specifically mention vehicles, vehicles are intended to be covered by the terms of “... use of... devices...” and “... occupancy of... structures or premises...” in Items 1 and 2 of the section. It is clear that the code

specifically intends to regulate vehicles because, in some cases, there are regulations in the code that are specific to vehicles, such as those in Sections 309, 4106, 5706 and 5707. However, in most cases, unless vehicles are specifically mentioned, provisions in the code would not apply to them (see also Section 904.13).

[A] 101.2.1 Appendices. Provisions in the appendices shall not apply unless specifically adopted.

C The provisions contained in Appendices A through N are not considered part of the code and are, therefore, not enforceable unless they are specifically included in the ordinance or other adopting law or regulation of the jurisdiction.

[A] 101.3 Purpose. The purpose of this code is to establish the minimum requirements for providing a reasonable level of life safety and property protection from the hazards of fire, explosion or dangerous conditions in new and existing buildings, structures and premises, and to provide a reasonable level of safety to firefighters and emergency responders during emergency operations.

C Code requirements regulate conditions that are likely to cause or contribute to fires or explosions, endanger life or property if a fire occurs or contribute to the spread of a fire. The code is intended to regulate conditions related to the health, safety and welfare of the public, firefighters and other emergency responders called on to conduct emergency operations in or on any building, structure or premises. Note that the code requirements are minimum (see commentary, Section 101.2 for a discussion on minimum requirements).

While the code does serve as a maintenance code for buildings constructed in accordance with the IBC, it has provisions that go far beyond maintenance of construction regulations for buildings. Much of the code prescribes construction regulations of several sorts, which can be seen throughout the code in the general provisions for safety and in special occupancies, processes and equipment. These regulations supplement the construction regulations in the IBC for cases where special hazards exist.

A common question that arises is, “Is it the intent of the code to apply to noncommercial structures?” Yes. The code applies to all structures within a jurisdiction, including residential occupancies, unless such occupancies are specifically excluded within the text of a particular code section. For example, Sections 503.1.1 and 507.1 require that all occupancies, including residential occupancies, be provided with fire apparatus access and a water supply for firefighting. Though it is true that there are many provisions in the code that would not normally be applicable to a residential occupancy based on the scope of the particular provision, the overall application of the code is not limited to commercial structures (see the commentaries to Section 503.1.1 and Appendix D107.1 for further discussion of this topic).

[A] 101.4 Severability. If a section, subsection, sentence, clause or phrase of this code is, for any reason, held to be unconstitutional, such decision shall not affect the validity of the remaining portions of this code.

C Only invalid sections of the code (as established by the court of jurisdiction) can be set aside. This is essential to safeguard the application of the code text to situations whereby a provision of the code is declared illegal or unconstitutional. This section preserves the legislative action that put the legal provisions in place.

[A] 101.5 Validity. In the event any part or provision of this code is held to be illegal or void, this shall not have the effect of making void or illegal any of the other parts or provisions hereof, which are determined to be legal; and it shall be presumed that this code would have been adopted without such illegal or invalid parts or provisions.

C The code provisions are intended to be construed as severable. If any part of the code is ruled invalid by a court of competent jurisdiction, the remaining sections of the code are intended to stand as though the invalid section never existed. Fire code officials and adopting bodies should carefully and promptly evaluate the impact of any such ruling on ongoing enforcement activities and the remaining code provisions. Such changes that are necessary to preserve and protect the enforcement authority of the jurisdiction and the public should be instituted through legislative action as soon as practical. Additionally, the International Code Council® (ICC®) should be advised of court actions invalidating any code provisions. For the same reason local officials must evaluate the effects of court decisions, the influence of court decisions on the remainder of the code must be evaluated for national impact.

SECTION 102—APPLICABILITY

[A] 102.1 Construction and design provisions. The construction and design provisions of this code shall apply to:

1. Structures, facilities and conditions arising after the adoption of this code.
2. Existing structures, facilities and conditions not legally in existence at the time of adoption of this code.
3. Existing structures, facilities and conditions where required in Chapter 11.
4. Existing structures, facilities and conditions that, in the opinion of the *fire code official*, constitute a distinct hazard to life or property.

C This section establishes the scope of application of the code provisions that regulate construction and design. Construction and design requirements include, but are not limited to, the installation of fire protection systems; drainage and secondary containment facilities for hazardous materials; fire-resistive construction and the activities stated in Section 105.7 for which a construction permit is required.

Item 1 specifies that construction and design code requirements apply to new construction that occurs following the adoption of the code.

Item 2 means that construction and design code requirements are to apply to existing structures that did not have a certificate of occupancy at the time the code was adopted. An example would be a building that was built when there was no adopted construction code in the jurisdiction.

Item 3 refers to Chapter 11, “Construction Requirements for Existing Buildings.” This chapter assembles in a single location all of the construction and design code requirements that specifically target existing structures, facilities and conditions for retroactive application.

Item 4 generally requires the fire code official to determine that a “distinct hazard to life or property” exists prior to enforcing a construction and design code provision retroactively. Simply claiming that a violation exists because a building does not comply with the most recent edition of the code does not necessarily establish that a hazard actually exists. The fire code official should be prepared to demonstrate, based on evidence or case histories that would be defensible in a court of law, that a distinct hazard exists. This would be especially true where enforcement would result in substantial expense to the property owner or when a building has remained in compliance with the edition of the code under which it was originally constructed.

Further legal counsel should be consulted prior to the retroactive application of the code in order to establish the defensibility of the fire code official’s determination in a court of law. Also, similar occupancies in the jurisdiction should be treated equally using a written policy to avoid the possibility of charges of selective enforcement.

[A] 102.2 Administrative, operational and maintenance provisions. The administrative, operational and maintenance provisions of this code shall apply to:

1. Conditions and operations arising after the adoption of this code.
2. Existing conditions and operations.

[C] This section specifies that the administrative, operational and maintenance requirements of the code apply to conditions and operations that exist when the code is adopted and new conditions and operations that begin after the code is adopted. Although Sections 102.1 and 102.2 are the controlling sections for retroactive application of the code to existing buildings, they do not provide for retroactive code application solely on the basis of a change in ownership or the occupying tenant. If a change in the occupancy group or the character of use occurs, the current edition of the code becomes enforceable. In such cases, Section 102.3 also applies.

[A] 102.3 Change of use or occupancy. A change of occupancy shall not be made unless the use or occupancy is made to comply with the requirements of this code and the *International Existing Building Code*.

Exception: Where *approved* by the *fire code official*, a change of occupancy shall be permitted without complying with the requirements of this code and the *International Existing Building Code*, provided that the new or proposed use or occupancy is less hazardous, based on life and fire risk, than the existing use or occupancy.

[C] A change of occupancy in an existing structure may change the level of inherent hazards, and a different code section may be applicable to the new occupancy. This section addresses a change of occupancy from one group to another or a change in the level of use of an occupancy; for example, a Group A-2 restaurant being changed to a nightclub. Another example relative to the code would be a Group S-1 warehouse that is adding high-piled combustible storage or changing the classification to a higher hazard storage commodity.

This section requires that the building comply with this code and the *International Existing Building Code*® (IEBC®). Three different options for compliance are provided within the IEBC: the prescriptive method, the work area method or the performance method. In the 2012 IBC, Chapter 34 addressed existing buildings and this code referred to the IBC for change of occupancy. This criterion from IBC Chapter 34 was repeated in the IEBC as Chapter 4, Prescriptive Compliance Methods, and Chapter 14 (what is now chapter 13), Performance Compliance Methods. Now this information is available only in the IEBC.

The exception acknowledges that compliance with this code and the IEBC is not necessary where the proposed use or occupancy would decrease the hazard level; for example, where a Group A-2 nightclub is changed to a Group A-2 restaurant. Note that the IEBC may need to be reviewed due to possible compliance with repair requirements or any alterations that are being addressed. In addition, the IEBC may be administered by another entity, typically the building department.

[A] 102.4 Application of building code. The design and construction of new structures shall comply with the *International Building Code*, and any *alterations*, additions, changes in use or changes in structures required by this code, which are within the scope of the *International Building Code*, shall be made in accordance therewith.

[C] The code is the companion fire and life safety maintenance code to the IBC. Maintenance of other building features is governed by other International Codes® (I-Codes®). When existing buildings change occupancy group or are altered, increased in area or demolished, the IBC provisions must apply. When compliance with the code requires alterations, additions or modifications within the scope of the IBC, the IBC regulations and the building official’s authority must prevail. This makes

it essential that the code officials responsible for enforcing the building and fire codes establish a sound working relationship. Clear communication is essential to achieve compliance with the respective code officials' orders.

[A] 102.5 Application of residential code. Where structures are designed and constructed in accordance with the *International Residential Code*, the provisions of this code shall apply as follows:

1. Construction and design provisions of this code pertaining to the exterior of the structure shall apply including, but not limited to, premises identification, fire apparatus access and water supplies. Where interior or exterior systems or devices are installed, construction permits required by Section 105.6 shall apply.
2. Administrative, operational and maintenance provisions of this code shall apply.

C This section clarifies the extent to which the *International Residential Code*® (IRC®) and the code are interrelated and how the provisions of the code apply to the development of one- and two-family dwelling projects built under the IRC.

The IRC is designed and intended for use as a stand-alone code for the construction of detached one- and two-family dwellings and townhouses not more than three stories in height. As such, the construction of detached one- and two-family dwellings and townhouses is regulated exclusively by the IRC and not subject to the provisions of any other I-Codes other than to the extent specifically referenced. Although the IRC regulates the construction of detached one- and two-family dwellings and townhouse structures, it does not regulate the design and construction of emergency access to and community fire protection for residential developments containing such dwelling structures. Accordingly, where the code is adopted, the design, construction, regulation and maintenance of fire apparatus access roads for servicing such residential developments must comply with the provisions of Section 503 and, if adopted, Appendix D. Also, the design, construction, regulation and maintenance of fire protection water supplies for servicing such residential developments must comply with the provisions of Section 507 and, if adopted, Appendices B and C. These specific requirements of the code are applicable because they include design and construction regulations that provide necessary emergency access and community fire protection for residential developments containing structures that are regulated within the scope of the IRC.

[A] 102.6 Historic buildings. The provisions of this code relating to the construction, *alteration*, repair, enlargement, restoration, relocation or moving of buildings or structures shall not be mandatory for existing buildings or structures identified and classified by the state or local jurisdiction as historic buildings where such buildings or structures do not constitute a distinct hazard to life or property. Fire protection in designated historic buildings shall be provided with an *approved* fire protection plan as required in Section 1103.1.1.

C This section gives the code official the widest possible flexibility in enforcing the code where the building in question has historic value. This flexibility, however, is not without conditions. The most important criterion for application of this section is that the building must be specifically classified as being of historic significance by a qualified party or agency. Usually, this is done by a state or local authority after considerable scrutiny of the historical value of the building. Most, if not all, states have such authorities, as do many local jurisdictions. The agencies with such authority can be located at the state or local government level or through the local chapter of the American Institute of Architects (AIA). This section requires an approved fire protection plan. More specifically, reference is made to Section 1103.1.1, which provides some minimal requirements. Section 1103.1.1 requires a plan to be developed in accordance with NFPA 914. The reference to NFPA 914 provides specific guidance to code officials, design professionals and building owners as to the provisions that are applicable to fire protection plans for historic buildings in order to provide a reasonable level of building and life safety provisions. Note that the IEBC also addresses historic buildings undergoing repairs, alterations and changes of occupancy.

[A] 102.7 Referenced codes and standards. The codes and standards referenced in this code shall be those that are listed in Chapter 80, and such codes and standards shall be considered to be part of the requirements of this code to the prescribed extent of each such reference and as further regulated by Sections 102.7.1 and 102.7.2.

C The use of referenced codes and standards to cover certain aspects of various occupancies and operations rather than write parallel or competing requirements into the code is a long-standing and successful code development principle. Often, however, questions and potential conflicts in the use of referenced codes and standards can arise, which can lead to inconsistent enforcement of the code. This section establishes the relationship between the code and the codes or standards that it references. A referenced code or standard or portion thereof is to be considered an enforceable extension of the code as if the specified content of the referenced code or standard were included in the body of the code. The extent to which the provisions of a referenced standard may be enforced is limited to those portions of the standard that are specifically identified in the code section that makes the reference. As an example of such limiting references, in Chapter 57, "Flammable and Combustible Liquids," the question has been posed as to whether the entire referenced standard, NFPA 30, is applicable since it is referenced 30 times in Chapter 57. The answer is no. The applicability of NFPA 30 content would be limited to only the specific content indicated in the code section making the reference. For example, Section 5703.6.2 limits the applicable NFPA 30 content to only Chapter 27 of that document; Section 5704.2.7 limits the applicable NFPA 30 content to only the tank design, fabrication and construction provisions of Chapters 21 and 22 or 23 of that document; and various other code sections, such as Sections 5704.2.7.8 and 5704.2.7.9, limit the reference to specifically enumerated sections of NFPA 30.

[A] 102.7.1 Conflicts. Where conflicts occur between provisions of this code and referenced codes and standards, the provisions of this code shall apply.

[C] The use of referenced codes and standards to cover certain aspects of various occupancies and operations, rather than write parallel or competing requirements into the code, is a longstanding code development principle. Often, however, questions and potential conflicts in the use of referenced codes and standards can arise, which can lead to inconsistent enforcement of the code. See the commentary to Section 102.7.2 for further discussion of conflicting provisions.

[A] 102.7.2 Provisions in referenced codes and standards. Where the extent of the reference to a referenced code or standard includes subject matter that is within the scope of this code, the provisions of this code, as applicable, shall take precedence over the provisions in the referenced code or standard.

[C] This section expands on the provisions of Section 102.7.1 by making it clear that, even if a referenced standard contains requirements that parallel the requirements of this code, the provisions of the code will always take precedence. One of the most common examples of such conflicting provisions is that many referenced standards contain building construction requirements that may differ from the requirements of the IBC, the applicability of which is established in Section 102.4 of the code. In such cases, the IBC would supersede the standard.

[A] 102.8 Subjects not regulated by this code. Where applicable standards or requirements are not set forth in this code, or are contained within other laws, codes, regulations, ordinances or bylaws adopted by the jurisdiction, compliance with applicable standards of the National Fire Protection Association or other nationally recognized fire safety standards, as *approved*, shall be deemed as prima facie evidence of compliance with the intent of this code. Nothing herein shall derogate from the authority of the *fire code official* to determine compliance with codes or standards for those activities or installations within the *fire code official's* jurisdiction or responsibility.

[C] This section provides guidance for situations in which no specific standard is designated in the code or otherwise adopted by the jurisdiction. In this instance, compliance with the requirements of a standard of the NFPA or other nationally recognized standards can be approved by the fire code official.

[A] 102.9 Matters not provided for. Requirements that are essential for the public safety of an existing or proposed activity, building or structure, or for the safety of the occupants thereof, that are not specifically provided for by this code, shall be determined by the *fire code official*.

[C] Evolving technology in our society will sometimes result in a situation or circumstance that the code does not cover. The reasonable application of the code to such hazardous, unforeseen conditions is provided in this section. Clearly, such a section is needed, and the fire code official's experience and judgment must be used. The section, however, does not override requirements that may be preferred when the code provides alternative methods. Additionally, the section can be used to implement the general performance-oriented language of the code in specific enforcement situations.

[A] 102.10 Conflicting provisions. Where there is a conflict between a general requirement and a specific requirement, the specific requirement shall be applicable. Where, in a specific case, different sections of this code specify different materials, methods of construction or other requirements, the most restrictive shall govern.

[C] The provisions of this section provide guidance to both fire code officials and other code users on the application of the code when different sections specify different materials, methods of construction or other requirements.

The importance of this section should not be understated. It resolves the question of how to handle conflicts between general and specific provisions found in the code, and those instances where different sections specify different requirements. This section provides a necessary hierarchy for application of code provisions and clarifies code applications that would otherwise leave persistent questions and lead to debate. The code requires that where different sections of the code apply but contain different requirements, the most restrictive provisions shall govern. The code also resolves conflicts between the general requirements of any particular issue with any specific requirements of the same issue by indicating that the specific requirements take precedence over the general requirements.

[A] 102.11 Other laws. The provisions of this code shall not be deemed to nullify any provisions of local, state or federal law.

[C] In some cases, other laws enacted by the jurisdiction or the state or federal government may be applicable to a condition that is also governed by a requirement in the code. In such circumstances, the requirements of the code are in addition to the existing law still in effect, although the code official may not be responsible for its enforcement.

[A] 102.12 Application of references. References to chapter or section numbers, or to provisions not specifically identified by number, shall be construed to refer to such chapter, section or provision of this code.

[C] In a situation where the code makes reference to a chapter or section number or to another code provision without specifically identifying its location in the code, assume that the referenced section, chapter or provision is in the code and not in a referenced code or standard.

PART 2—ADMINISTRATION AND ENFORCEMENT**SECTION 103—CODE COMPLIANCE AGENCY**

[A] 103.1 Creation of agency. The [INSERT NAME OF DEPARTMENT] is hereby created and the official in charge thereof shall be known as the *fire code official*. The function of the agency shall be the implementation, administration and enforcement of the provisions of this code.

C The traditional enforcement agency for the code is the fire department or fire prevention bureau of a state, county or municipal government. Such agencies usually perform administrative functions and provide public safety services related to fire protection; however, a variety of less-traditional arrangements have also been used to enforce the code, including private corporations, such as fire districts and fire companies employed by a local government to act as its agent; police and other law enforcement agencies; building, housing or zoning authorities; and community and economic development departments. Regardless of who is designated by the legislative or administrative authority to adopt and enforce the code, this section establishes the legal duty of the fire code official to enforce the code.

[A] 103.2 Appointment. The *fire code official* shall be appointed by the chief appointing authority of the jurisdiction.

C A code official's independence is essential so that public safety decisions are not based on political, economic or social expediency. This is not to say that social, political and economic considerations should not matter when deciding some code questions, but the interests of public health, safety and welfare must not be compromised to achieve such objectives. Protection of code officials from removal from office without cause helps ensure that reasonable and competent professionals will be willing to serve.

[A] 103.3 Deputies. In accordance with the prescribed procedures of this jurisdiction and with the concurrence of the appointing authority, the *fire code official* shall have the authority to appoint a deputy *fire code official*, other related technical officers, inspectors and other employees. Such employees shall have powers as delegated by the *fire code official*.

C Most jurisdictions require more than one person to enforce the code. While there is only one code official, there can be any number of deputies working under the direction of the code official. With the technical and legal demands on code enforcers increasing, additional personnel will certainly be required in this area to serve adequately the public interest. Though the professional qualifications of code officials are not detailed in the code, individuals appointed to code enforcement positions should be technically competent, motivated, and well-adapted and possess good written and oral communication skills.

SECTION 104—DUTIES AND POWERS OF THE FIRE CODE OFFICIAL

[A] 104.1 General. The *fire code official* is hereby authorized to enforce the provisions of this code.

C The duty of the fire code official is to enforce the code, and they are the "authority having jurisdiction" for all matters relating to the code and its enforcement.

The provisions in Section 104 and its subsections are found in all the I-Codes, and with a few minor exceptions are identical in all the I-Codes. This provides the code official enforcing multiple codes with the ability to address application and enforcement of all those codes in the same manner.

[A] 104.2 Determination of compliance. The *fire code official* shall have the authority to determine compliance with this code, to render interpretations of this code and to adopt policies and procedures in order to clarify the application of its provisions. Such interpretations, policies and procedures:

1. Shall be in compliance with the intent and purpose of this code.
2. Shall not have the effect of waiving requirements specifically provided for in this code.

C It is the duty of the fire code official to interpret the code and to determine compliance. Code compliance will not always be easy to determine and will require judgment and expertise, particularly when enforcing the provisions of Sections 104.2.3 and 104.2.4.

The code provides the fire code official with several avenues to determine compliance with the code, the first being strict compliance with the code provisions. This provides the simplest route. Second is reliance on listed products or materials. The listing demonstrates to the fire code official that the product has been evaluated and complies with the requirements in the code. Third is alternative methods and materials, a route that allows the applicant to suggest an alternative path to compliance with the intent of the code. Finally, the fire code official can grant a modification. This option is applicable in situations where compliance with the code is not practical and allows the fire code official to modify the requirements.

The fire code official is authorized to make interpretations of the code requirements and develop policies consistent with the intent of the code. In exercising this authority, however, the fire code official cannot set aside or ignore any provision of the code. Frequently, interpretations become policies that are applicable to all new construction. These policies must be in writing and set down the official interpretation of the jurisdiction. Interpretations and policies do not create new requirements, but rather provide guidance and demonstrate what is acceptable where the code simply states, "as approved by the fire code official."

[A] 104.2.1 Listed compliance. Where this code or a referenced standard requires equipment, materials, products or services to be listed and a listing standard is specified, the listing shall be based on the specified standard. Where a listing standard is not specified, the listing shall be based on an approved listing criteria. Listings shall be germane to the provision requiring the listing. Installation shall be in accordance with the listing and the manufacturer's instructions, and where required to verify compliance, the listing standard and manufacturer's instructions shall be made available to the *fire code official*.

[C] The code requires listed equipment in many situations. Where a listing is required, the listing must be in accordance with the referenced standard. It is critical that the listing evaluates the intended use and application of the product. In cases where the code specifies the listing standard, typically the fire code official will accept products listed to that standard without further evaluating whether the standard is germane to the use. When a product listing is appropriate, the fact that the product is listed and then installed in accordance with the listing specifications and the manufacturer's instructions becomes the approval of the product. However, where the code does not provide a specific listing standard, the appropriate listing standard shall be determined by the fire code official.

[A] 104.2.2 Technical assistance. To determine compliance with this code, the *fire code official* is authorized to require the owner or owner's authorized agent to provide a technical opinion and report.

[C] It is unlikely that one person has the technical knowledge to evaluate all operations and uses from a safety standpoint. This section authorizes the fire code official to require the owner or owner's authorized agent to provide a technical opinion safety report. The report is to be prepared by parties that have the technical ability to evaluate the design of the facility or the operational process in question. A registered design professional is commonly used for these services. It is critical that the preparer of the report have the proper background and experience for the project, since the credibility of the report depends on these qualifications.

[A] 104.2.2.1 Cost. A technical opinion and report shall be provided without charge to the jurisdiction.

[C] The cost for preparation of the technical opinion and report is paid by the applicant. Tests to demonstrate compliance with the code can also be submitted, provided that they are conducted by a third party approved by the building official.

[A] 104.2.2.2 Preparer qualifications. The technical opinion and report shall be prepared by a qualified engineer, specialist, laboratory or fire safety specialty organization acceptable to the *fire code official*. The *fire code official* is authorized to require design submittals to be prepared by, and bear the stamp of, a *registered design professional*.

[C] This section requires that the person or agency providing the technical report must be qualified. The fire code official can require that the report is stamped by a registered design professional, since not all reports may need to provide this. For example, a hazardous materials classification report often does not include engineering or design. Also, the report may be a source document, as opposed to a review of documentation prepared by others.

[A] 104.2.2.3 Content. The technical opinion and report shall analyze the properties of the design, operation or use of the building or premises and the facilities and appurtenances situated thereon to identify and propose necessary recommendations.

[C] The technical report shall include an analysis and any recommended or necessary changes.

[A] 104.2.2.4 Tests. Where there is insufficient evidence of compliance with the provisions of this code, the *fire code official* shall have the authority to require tests as evidence of compliance. Test methods shall be as specified in this code or by other recognized test standards. In the absence of recognized test standards, the *fire code official* shall approve the testing procedures. Such tests shall be performed by a party acceptable to the *fire code official*.

[C] To provide the basis on which the fire code official can make a decision regarding an alternative material or method, sufficient technical data, test reports and documentation must be provided for evaluation. If evidence satisfactory to the fire code official proves that the alternative material or construction method is equivalent to that required by the code, they may approve it. Any such approval cannot have the effect of waiving any requirements of the code. The burden of proof of equivalence lies with the applicant who proposes the use of alternative materials or methods.

The fire code official must require the submission of any appropriate information and data to assist in the determination of equivalency. This information must be submitted before a permit can be issued. The type of information required includes test data in accordance with referenced standards, evidence of compliance with the referenced standard specifications and design calculations. If no test standard is available, the owner or owner's agent is to propose the test procedures and submit them to the fire code official for review and approval. For example, ICC-ES has issued evaluation reports that would be particularly useful in providing the fire code official with the technical basis for evaluation and approval of new and innovative materials and components. The use of such authoritative and credible research reports can greatly assist the fire code official by reducing the time-consuming engineering analysis necessary to review materials and products. Failure to adequately substantiate a request for the use of an alternative is a valid reason for the fire code official to deny a request.

[A] 104.2.3 Alternative materials, design and methods of construction and equipment. The provisions of this code are not intended to prevent the installation of any material or to prohibit any design or method of construction not specifically prescribed by this code, provided that any such alternative is not specifically prohibited by this code and has been *approved*.

Exception: Performance-based alternative materials, designs or methods of construction and equipment complying with the *International Code Council Performance Code*.

C The code is a compilation of criteria with which materials, equipment, devices and systems must comply to be acceptable for a particular application. The code provides methods to mitigate hazards and provide for safe operations, but alternative methods or materials may provide an equivalent level of safety. The owner or designer has the option to submit alternative methods for approval, and the fire code official has a duty to evaluate such materials, equipment, devices and systems for code compliance. Approval of any alternative method or material must be with the concurrence of the building and fire code official. As a result of this approval, the material, equipment, device or system must be constructed and installed in compliance with that approval, and with all the conditions and limitations considered as a basis for that approval.

This process ensures due process of law and also allows for appeal, covered in Section 111. Similar language is found in all of the I-Codes.

The exception is intended to make it clear that the *International Code Council Performance Code*® (ICCPC®) does not have to be considered solely for whole-building designs. It can also be another pathway for evaluating alternative materials, designs and methods of construction. When projects are designed to the prescriptive requirements of any I-Code, there are situations where a single material, element, or system cannot conform to the prescriptive requirements. Also, new materials, elements or systems are entering the construction market at a pace such that the prescriptive codes cannot keep up. This provision allows owners, designers and building officials to consider such advances in materials and elements of designs using the ICCPC for guidance. Although the prescriptive provisions in each of the codes provides one pathway for approval of alternative materials, designs and methods of construction, the ICCPC should not be overlooked as an alternative pathway.

[A] 104.2.3.1 Approval authority. An alternative material, design or method of construction shall be *approved* where the *fire code official* finds that the proposed alternative is satisfactory and complies with Sections 104.2.3.2 through 104.2.3.7, as applicable.

C This section makes it clear that if the alternative is acceptable in accordance with the following noted subsections, it can be approved by the fire code official.

[A] 104.2.3.2 Application and disposition. Where required, a request to use an alternative material, design or method of construction shall be submitted in writing to the *fire code official* for approval. Where the alternative material, design or method of construction is not *approved*, the *fire code official* shall respond in writing, stating the reasons the alternative was not *approved*.

C This section requires that the submittal for an alternative must be in writing. If it is not approved, the fire code official must so state in writing and provide reasons why it was not acceptable. This allows the applicant the ability to modify the request or provide additional documentation.

[A] 104.2.3.3 Compliance with code intent. An alternative material, design or method of construction shall comply with the intent of the provisions of this code.

C This section makes it clear that although the use of the proposed alternative material, design or method of construction is not prescriptively addressed in the code, it still must comply with the intent of the code.

[A] 104.2.3.4 Equivalency criteria. An alternative material, design or method of construction shall, for the purpose intended, be not less than the equivalent of that prescribed in this code with respect to all of the following, as applicable:

1. Quality.
2. Strength.
3. Effectiveness.
4. Durability.
5. Safety, other than fire safety.
6. Fire safety.

C This section requires that the alternative must provide equivalency to the code's provisions. The list of characteristics to be addressed is included to assist in the approval. Any such approval cannot have the effect of waiving any requirements of the code and must demonstrate equivalency to the code's intent with regard quality, strength, effectiveness, durability, fire safety, and safety other than fire safety. Fire safety is separated from the characteristic of other safety concerns. Previously, the code only required equivalency to fire resistance, but fire safety expands that to include other aspects such as performance under fire conditions. The characteristic of safety would include other aspects of safety such as egress. The responsibility to demonstrate equivalency lies with the applicant who proposes the use of alternative materials or methods.

The applicant must submit the appropriate information and data to assist in the determination of equivalency. This information must be submitted in writing and reviewed before a permit can be issued. The type of information required includes test data in accordance with referenced standards, evidence of compliance with the referenced standard specifications and design calculations.

[A] 104.2.3.5 Tests. Tests conducted to demonstrate equivalency in support of an alternative material, design or method of construction application shall be of a scale that is sufficient to predict performance of the end use configuration. Tests shall be performed by a party acceptable to the *fire code official*.

[C] This section makes it clear that it is required that the size of the testing conducted is to be performed to a scale that adequately represents the end use of the alternative.

[A] 104.2.3.5.1 Fire tests. Tests conducted to demonstrate equivalent fire safety in support of an alternative material, design or method of construction application shall be of a scale that is sufficient to predict fire safety performance of the end use configuration. Tests shall be performed by a party acceptable to the *fire code official*.

[C] The intent of this section is to specifically clarify equivalency in terms of fire safety. A proper fire safety analysis performed under this section should always take these considerations into account but having them specifically stated separately helps fire code officials and code users by providing more thorough guidance for preparation of alternative method proposals. Additional guidance is also provided to ensure that the fire testing done in support of an alternative method proposal is of a sufficient scale to be relevant to the end use application.

[A] 104.2.3.6 Reports. Supporting data, where necessary to assist in the approval of materials or assemblies not specifically provided for in this code, shall comply with Sections 104.2.3.6.1 and 104.2.3.6.2.

[C] Where an alternative material or method is proposed for construction, it is incumbent on the fire code official to determine whether the alternative is, in fact, equivalent to the methods prescribed by the code.

[A] 104.2.3.6.1 Evaluation reports. Evaluation reports shall be issued by an *approved agency* and use of the evaluation report shall require approval by the *fire code official* for the installation. The alternate material, design or method of construction and product evaluated shall be within the scope of the *fire code official's* recognition of the *approved agency*. Criteria used for the evaluation shall be identified within the report and, where required, provided to the *fire code official*.

[C] Reports providing evidence of this equivalency are required to be supplied by an approved source, meaning a source that the fire code official finds to be reliable and accurate. The ICC Evaluation Service (ICC-ES) is an example of an agency that provides research reports for alternative materials and methods.

This evaluation report is conducted by an approved agency that is accredited to conduct the tests or evaluations appropriate for the alternative involved. When the applicant provides a product evaluation from an accredited product evaluation agency that uses publicly developed and available criteria for the evaluation, the code official may have increased confidence that the method used for the evaluation does result in a method or material that meets the intent of the code and is at least equivalent to code-prescribed construction.

Public development of criteria allows for input from industry experts, the public and building officials in determining the methods used to evaluate code intent and equivalence. This is similar to the code development process, where consensus is important. The accreditation ensures that the organization uses a consistent process to perform evaluations. This section is meant to reflect the current use of evaluation reports from accredited evaluation agencies or organizations.

[A] 104.2.3.6.2 Other reports. Reports not complying with Section 104.2.3.6.1 shall describe criteria, including but not limited to any referenced testing or analysis, used to determine compliance with code intent and justify code equivalence. The report shall be prepared by a qualified engineer, specialist, laboratory or fire safety specialty organization acceptable to the *fire code official*. The *fire code official* is authorized to require design submittals to be prepared by, and bear the stamp of, a *registered design professional*.

[C] This section is intended to address reports generated by persons or agencies other than an approved agency. It specifies that the person or agency providing the report must be qualified and must be approved by the fire code official. The fire code official has the authority to require the stamp of a registered design professional. When an applicant provides an evaluation from other than an accredited agency, or from a source that does not use publicly developed and available criteria, the fire code official needs more information in order to perform a proper review. Not only does the fire code official need to evaluate the product, but also evaluate the method that the applicant has used to determine compliance with code intent and code equivalence. In such cases, it is proposed that the applicant would also have to provide the criteria that was used to do the evaluation, justification for use of that criteria and data used for the evaluation so a complete review can be made.

[A] 104.2.3.7 Peer review. The *fire code official* is authorized to require submittal of a *peer review* report in conjunction with a request to use an alternative material, design or method of construction, prepared by a peer reviewer that is *approved* by the *fire code official*.

[C] This section addresses a method of review that is utilized by many jurisdictions. The peer review is an outside, third-party review that is submitted to the code official for use in cases where a jurisdiction may not have qualified resources in-house

to perform a sufficient review of an alternative compliance proposal. Again, the peer reviewer must be qualified and approved by the fire code official.

104.2.4 Modifications. Where there are practical difficulties involved in carrying out the provisions of this code, the *fire code official* shall have the authority to grant modifications in accordance with Section 104.2.4.1 or 104.2.4.2.

C The fire code official may amend or make exceptions to the code as needed where strict compliance is impractical. Only the fire code official has authority to grant modifications. Consideration of a particular difficulty is to be based on the application of the owner and a demonstration that the purpose of the code is accomplished. This section is not intended to permit setting aside or ignoring a code provision; rather, it is intended to provide acceptance of equivalent protection. Such modifications do not, however, extend to actions that are necessary to correct violations of the code. In other words, a code violation or the expense of correcting one cannot constitute a practical difficulty.

Filing the details of a modification action is necessary if the reasons for the modification are subject to review. Comprehensive written records are an essential part of an effective administrative system. Unless clearly written records of the considerations and documentation utilized in the modification process are created and maintained, subsequent enforcement action will be difficult to support and will be inconsistent.

104.2.4.1 Individual cases. The *fire code official* shall have the authority to grant modifications for individual cases, provided that the *fire code official* shall first find that special individual reason makes the strict letter of this code impractical and the modification is in compliance with the intent and purpose of this code and that such modification does not lessen health, life and fire safety requirements. The details of action granting modifications shall be recorded and entered in the files of the code compliance agency.

C This section and Section 104.2.4.2 provide the need for granting authority to the fire code official to allow, by policy, conditions that would otherwise constitute code violations. For example, long-term power outages will eventually render many alarm systems nonfunctional, and extended loss of heat in buildings can lead to catastrophic freezing of fire suppression systems. Shutting down such systems and draining them can prevent catastrophic damage, allowing a system that might otherwise take months to repair to be placed back into service more quickly. If water remains in a system and freezing occurs, the system is nonfunctional anyway, so whether drained or not, protection is going to be impaired for some period of time. But, allowing more of a system to freeze vs. draining can be expected to result in increased water damage when the system thaws and much more extensive and time-consuming repairs. This section could also be used to allow temporary emergency shelters that may not fully meet code requirements for a congregate residential use.

104.2.4.2 Natural disasters. In preparation for, during and after a natural disaster event, as determined by the *fire code official*, the *fire code official* shall have the authority to issue written policies, procedures or rules that modify this code as necessary to protect life and property. Such policies, procedures or rules shall be made available to the public and shall include start and end dates, which can be extended at the *fire code official's* discretion.

C This section authorizes the fire code official to develop written documentation globally addressing special allowances permitted during a disaster event. As written, the section places the authority to make any such allowances solely in the hands of the fire code official. These provisions are only in this code because the IFC is unique among the I-Codes with respect to its application to operation of existing buildings and to emergency response.

[A] 104.3 Applications and permits. The *fire code official* is authorized to receive applications, review *construction documents* and issue permits for construction regulated by this code, issue permits for operations regulated by this code, inspect the premises for which such permits have been issued and enforce compliance with the provisions of this code.

C The fire code official is obligated to receive applications, review construction documents, issue permits, conduct inspections and enforce the provisions of the code. They are to provide the services required to carry the project from application for the permit to final approval. The fire code official is to accept all properly completed applications and not refuse the receipt of an application that meets the policy requirements. This same principle holds for the review of the construction documents, issuance of permits, inspections and enforcement of the code's provisions. The requirements of the code must be met and approval will be granted only when compliance is verified.

[A] 104.4 Right of entry. Where it is necessary to make an inspection to enforce the provisions of this code, or where the *fire code official* has reasonable cause to believe that there exists in a structure or on any premises a condition that is contrary to or in violation of this code that makes the structure or premises unsafe, dangerous or hazardous, the *fire code official* is authorized to enter the structure or premises at all reasonable times to inspect or to perform the duties imposed on the *fire code official* by this code. If such structure or premises is occupied, the *fire code official* shall present credentials to the occupant and request entry. If such structure or premises is unoccupied, the *fire code official* shall first make a reasonable effort to locate the *owner*, the *owner's* authorized agent or other person having charge or control of the structure or premises and request entry. If entry is refused, the *fire code official* shall have recourse to every remedy provided by law to secure entry.

C This section establishes the right of the fire code official to enter the premises to make the inspections required by Section 105.2.2. The right to enter structures or premises is limited for a variety of reasons.

First, to protect the right of privacy, the owner or occupant must grant the code official permission before an interior inspection of the property can be conducted. Permission is not required for inspections that can be accomplished from within the public right-of-way.

Second, such access may be denied by the owner or occupant. Unless the inspector has reasonable cause to believe that a code violation exists, access may be unattainable. Searches of a private residence to gather information for the purpose of enforcing codes, ordinances or regulations are considered unreasonable and are prohibited by the Fourth Amendment to the US Constitution. “Reasonable cause” in the context of this section must be distinguished from “probable cause,” which is required to gain access to property in criminal cases. The burden of proof for establishing reasonable cause may vary among jurisdictions. Usually, an inspector must show that the property is subject to inspection under the provisions of the code; that the interests of the public health, safety and welfare outweigh the individual’s right to maintain privacy; and that such an inspection is required solely to determine compliance with the provisions of the code.

Many jurisdictions do not recognize the concept of an administrative warrant and may require the code official to prove probable or reasonable cause in order to gain access upon refusal. This burden of proof is usually more substantial, often requiring the code official to stipulate in advance why access is needed (usually access is restricted to gathering evidence for seeking an indictment or making an arrest); what specific items or pieces of information are sought; its relevance to the case against the individual subject; how knowledge of the relevance of the information or items sought was obtained; and how the evidence sought will be used. In all such cases, the right to privacy must always be weighed against the right of the code official to conduct an inspection to verify that the public health, safety and welfare are not in jeopardy. Such important and complex constitutional issues should be discussed with the jurisdiction’s legal counsel. Jurisdictions should establish procedures for securing the necessary court orders when an inspection is deemed necessary following a refusal.

Third, code officials must present proper identification (see commentary, Section 104.5) and request admittance during reasonable hours—usually the normal business hours of the establishment—to be admitted.

Fourth, inspections must be aimed at securing or determining compliance with the provisions and intent of the regulations that are specifically within the established scope of the code official’s authority.

[A] 104.4.1 Warrant. Where the *fire code official* has first obtained a proper inspection warrant or other remedy provided by law to secure entry, an *owner*, the *owner’s* authorized agent, occupant or person having charge, care or control of the structure or premises shall not fail or neglect, after a proper request is made as herein provided, to permit entry therein by the *fire code official* for the purposes of inspection and examination pursuant to this code.

C Very simply, the requirements in this section specify that where the fire code official has obtained a warrant to inspect the property, the owner, owner’s authorized agent or occupant is to allow the fire code official entry to do the inspection (see commentary, Section 104.4).

[A] 104.5 Identification. The *fire code official* shall carry proper identification when inspecting structures or premises in the performance of duties under this code.

C This section requires the fire code official (including, by definition, all authorized designees) to carry identification when conducting the duties of the position. The identification removes any question of the purpose and authority of the inspector (see Commentary Figure 104.5).

Commentary Figure 104.5—Personnel Identification Badge

City of (Jurisdiction)
Fire Department

Name of individual

Job function

The individual identified on the badge is a duly authorized employee of the (Jurisdiction) and is a designated representative of the Fire Department.

Valid through _____ Date _____ Building official _____

[A] 104.6 Notices and orders. The *fire code official* shall issue necessary notices or orders to ensure compliance with this code. Notices of violations shall be in accordance with Section 113.

[C] The fire code official is required to issue orders to abate illegal or hazardous conditions and to pursue correction or abatement of hazardous conditions by issuing legal notices and orders as described by the code. Courts are increasingly ruling that failure to follow up and pursue appropriate legal remedies promptly exposes both the fire code official and the jurisdiction to a liability in tort.

[A] 104.7 Official records. The *fire code official* shall keep official records as required by Sections 104.7.1 through 104.7.6. Such official records shall be retained for not less than 5 years or for as long as the building or structure to which such records relate remains in existence, unless otherwise provided by other regulations.

[C] In keeping with the need for an efficiently conducted business practice, the fire code official must keep official records. Such documentation provides a valuable resource of information if questions arise throughout the life of the building and its occupants. The code requires that the construction documents be kept until the project is complete or for at least 5 years, whichever is longer.

[A] 104.7.1 Approvals. A record of approvals shall be maintained by the *fire code official* and shall be available for public inspection during business hours in accordance with applicable laws.

[C] Records of prior approvals may be needed to determine the status of an existing operation or for future validation of a specific condition. Some jurisdictions outsource the plan review process to a third party. The approved documents, even though approved by the third party, would be maintained by the fire code official. Where a technical report from an expert is provided as part of the review process, the fire code official shall retain that report as part of the approval documentation.

[A] 104.7.2 Inspections. The *fire code official* shall keep a record of each inspection made, including notices and orders issued, showing the findings and disposition of each.

[C] Records of inspections are needed to support the issuance of a certificate of occupancy. The inspection records should document any code violations that were subsequently corrected.

104.7.3 Fire records. The *fire code official* shall keep a record of fires occurring within its jurisdiction and of facts concerning the same, including statistics as to the extent of such fires and the damage caused thereby, together with other information as required by the *fire code official*.

[C] Fire records provide a history of the fire experience of a facility and a cumulative record for all of the facilities of a jurisdiction. Fire records support consideration for construction code requirements based on the need to prevent additional fire occurrences.

[A] 104.7.4 Code alternatives and modifications. Application for alternative materials, design and methods of construction and equipment in accordance with Section 104.2.3; modifications in accordance with Section 104.2.4; and documentation of the final decision of the *fire code official* for either shall be in writing and shall be retained in the official records.

[C] The fire code official may approve alternative methods and materials or code modifications. Filing the details of a modification action is necessary if the reasons for the modification are subject to review. Comprehensive written records are an essential part of an effective administrative system. Unless clearly written records of the considerations and documentation utilized in the modification process are created and maintained, subsequent enforcement action will be difficult to support and will be inconsistent.

[A] 104.7.5 Tests. The *fire code official* shall keep a record of tests conducted to comply with Sections 104.2.2.4 and 104.2.3.5.

[C] This section clarifies that these test records are required to be maintained by the fire code official to provide possible future evidence of compliance.

[A] 104.7.6 Fees. The *fire code official* shall keep a record of fees collected and refunded in accordance with Section 108.

[C] This section clarifies that the fee records are required to be maintained by the fire code official to provide possible future evidence of how much was paid and refunded. Fees can be charged for permits, plan review and inspection. Section 108 provides details on the fee process and includes receiving fees, refunding fees and establishing fees. The receipt and refund of fees must be recorded and available for review and account balancing.

[A] 104.8 Liability. The *fire code official*, member of the board of appeals, officer or employee charged with the enforcement of this code, while acting for the jurisdiction, in good faith and without malice in the discharge of the duties required by this code or other pertinent law or ordinance, shall not thereby be rendered personally liable, either civilly or criminally, and is hereby relieved from all personal liability for any damage accruing to persons or property as a result of any act or by reason of an act or omission in the discharge of official duties.

[C] The fire code official, other department employees and members of the appeals board are not intended to be held liable, civilly or criminally, for those actions performed in accordance with the code in a reasonable and lawful manner. However,

the responsibility of the fire code official in this regard is subject to local, state and federal laws that may supersede this provision. This section further establishes that fire code officials (or subordinates) must not be liable for costs in any legal action instituted in response to the performance of lawful duties. Section 104.8.1 states that those costs are to be borne by the state, county or municipality, as applicable. The best way to be certain that the fire code official's action is a "lawful duty" is always to cite the applicable code section on which the enforcement action is based.

[A] 104.8.1 Legal defense. Any suit or criminal complaint instituted against any officer or employee because of an act performed by that officer or employee in the lawful discharge of duties and under the provisions of this code or other laws or ordinances implemented through the enforcement of this code shall be defended by the legal representatives of the jurisdiction until the final termination of the proceedings. The *fire code official* or any subordinate shall not be liable for costs in an action, suit or proceeding that is instituted in pursuance of the provisions of this code; and any officer of the department of fire prevention, acting in good faith and without malice, shall be free from liability for acts performed under any of its provisions or by reason of any act or omission in the performance of official duties in connection therewith.

[C] This section establishes that fire code officials or subordinates must not be liable for costs in any legal action, whether criminal or civil in nature, in response to the performance of lawful duties. This section states that these costs must be borne by the state, county or municipality. The best way to be certain that the fire code official's action is a lawful duty is to always cite the applicable code section substantiating the action.

104.9 Approved materials and equipment. Materials, equipment and devices *approved* by the *fire code official* shall be constructed and installed in accordance with such approval.

[C] The code is a compilation of criteria with which materials, equipment, devices and systems must comply to be suitable for a particular application. The fire code official has a duty to evaluate such materials, equipment, devices and systems for code compliance and, when compliance is determined, approve the same for use. The materials, equipment, devices and systems must be constructed and installed in compliance with, and all conditions and limitations considered as a basis for, that approval. For example, the manufacturer's instructions and recommendations are to be followed if the approval of the material was based, even in part, on those instructions and recommendations.

The approval authority given to the fire code official is a significant responsibility and is a key to code compliance. The approval process is first technical and then administrative and must be approached as such. For example, if data to determine code compliance are required, such data should be in the form of test reports or engineering analyses and not simply taken from a sales brochure.

[A] 104.9.1 Materials and equipment reuse. Materials, equipment and devices shall not be reused unless such elements are in good working order and *approved*.

[C] Code criteria for materials and equipment have changed over the years. Evaluation of testing and materials technology has permitted the development of new criteria that old materials may not satisfy. As a result, used materials are required to be evaluated in the same manner as new materials. Used materials, equipment and devices must be equivalent to that required by the code if they are to be used again in a new installation.

104.10 Fire investigations. The *fire code official*, the fire department or other responsible authority shall have the authority to investigate the cause, origin and circumstances of any fire, explosion or other hazardous condition. Information that could be related to trade secrets or processes shall not be made part of the public record, except as directed by a court of law.

[C] The prompt and thorough investigation of fires is important for many reasons, not the least of which are the identification of incendiary fires and prosecution of arsonists. In such cases, the duty of the fire code official is clear evidence must be preserved and leads pursued through criminal prosecution, if possible. However, a more important and frequently overlooked aspect of fire investigation is loss analysis. Whether or not the fire code official has jurisdiction to investigate incendiary fires and prosecute arsonists, it is extremely important that the enforcement agency be involved in the process of determining why fires occur, what can be done to prevent fires, how their effects can be lessened and how people behave once fires occur. Such lessons gleaned from past tragedies have influenced the code-development efforts of various organizations across the country. The second sentence recognizes the sensitivity of trade secrets that may be involved. This section cautions that such information not be released unless permitted by the court.

104.10.1 Assistance from other agencies. Police and other enforcement agencies shall have authority to render necessary assistance in the investigation of fires when requested to do so.

[C] Where needed, the fire code official has the authority to ask for assistance from the police department or other enforcement agencies, such as fire code officials in nearby jurisdictions, to investigate fires.

104.11 Authority at fires and other emergencies. The *fire chief* or officer of the fire department in charge at the scene of a fire or other emergency involving the protection of life or property, or any part thereof, shall have the authority to direct such operation as necessary to extinguish or control any fire, perform any rescue operation, investigate the existence of suspected or reported fires, gas leaks or other hazardous conditions or situations, or take any other action necessary in the reasonable performance of duty. In the exercise of such power, the *fire chief* is authorized to prohibit any person, vehicle, vessel or thing from approaching the scene,

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and is authorized to remove, or cause to be removed or kept away from the scene, any vehicle, vessel or thing that could impede or interfere with the operations of the fire department and, in the judgment of the *fire chief*, any person not actually and usefully employed in the extinguishing of such fire or in the preservation of property in the vicinity thereof.

C The specific conditions of authority that are granted to the fire code official at a fire or other emergencies are described. The first half of the paragraph simply describes the fire code official's authority to carry out the fire operation at the site. The fire code official also needs to be able to control who and what are allowed to be at the site so that emergency operations are not hampered.

104.11.1 Barricades. The *fire chief* or officer of the fire department in charge at the scene of an emergency is authorized to place ropes, guards, barricades or other obstructions across any street, alley, place or private property in the vicinity of such operation so as to prevent accidents or interference with the lawful efforts of the fire department to manage and control the situation and to handle fire apparatus.

C The fire code official has the authority to control access to the emergency site so that firefighting operations can occur without interference. This authority is also addressed in Section 104.11.

104.11.2 Obstructing operations. Persons shall not obstruct the operations of the fire department in connection with extinguishment or control of any fire, or actions relative to other emergencies, or disobey any lawful command of the *fire chief* or officer of the fire department in charge of the emergency, or any part thereof, or any lawful order of a police officer assisting the fire department.

C Fire department operations cannot be obstructed and directions from the fire department official in command at the emergency site must be carried out. This is necessary for efficient emergency operations.

104.11.3 Systems and devices. Persons shall not render a system or device inoperative during an emergency unless by direction of the *fire chief* or fire department official in charge of the incident.

C This section is an extension of the requirements in Section 104.11. The fire department official is in complete charge of the firefighting operation at the site. No person is to tamper with the equipment needed for the emergency.

SECTION 105—PERMITS

[A] 105.1 General. Permits shall be in accordance with Sections 105.1.1 through 105.6.25.

C This section includes the regulations covering permits, including a comprehensive list of the kinds of activities that require permits.

[A] 105.1.1 Permits required. A property owner or owner's authorized agent who intends to conduct an operation or business, or install or modify systems and equipment that are regulated by this code, or to cause any such work to be performed, shall first make application to the *fire code official* and obtain the required permit.

C The property owner or an owner's authorized agent is required to make application for and obtain a permit. It is important that the owner or an owner's authorized agent performs this function to confirm that they are aware of and give consent for permitted operations that may include hazardous materials that could pollute or contaminate the property. See the commentary for permit fees (Section 108.2) and keeping permits on the premises and available for inspection (Section 105.3.5).

105.1.2 Types of permits. There shall be two types of permits as follows:

1. Operational permit. An operational permit allows the applicant to conduct an operation or a business for which a permit is required by Section 105.5 for either:
 - 1.1. A prescribed period.
 - 1.2. Until renewed or revoked.
2. Construction permit. A construction permit allows the applicant to install or modify systems and equipment for which a permit is required by Section 105.6.

C The types of activities that require an operational permit are listed in Section 105.5. Construction activities that require a permit are listed in Section 105.6.

105.1.3 Multiple permits for the same location. Where more than one permit is required for the same location, the *fire code official* is authorized to consolidate such permits into a single permit provided that each provision is listed in the permit.

C The code allows for a number of activities to be included on a single permit in order to decrease the paperwork for all concerned. In this instance, the permit must list in detail the activities that are covered by the combined permit.

[A] 105.1.4 Emergency repairs. Where equipment replacement and repairs must be performed in an emergency situation, the permit application shall be submitted within the next working business day to the *fire code official*.

C In some cases, emergency replacement and repair work must be done as quickly as possible, so it is not practical to take the necessary time to apply for and obtain approval. A permit for the work must be obtained the next day that the building department is open for business. Any work performed before the permit is issued must be done in accordance with the code and corrected if not approved by the fire code official. For example, if a concealed trap failed on a Sunday, the

plumber could replace the trap at that time, but he would have to apply for a permit on Monday and have the repair pass an inspection.

[A] 105.1.5 Repairs. Application or notice to the *fire code official* is not required for ordinary repairs to structures, equipment or systems. Such repairs shall not include the cutting away of any wall, partition or portion thereof, the removal or change of any required *means of egress*, or rearrangement of parts of a structure affecting the egress requirements; nor shall any repairs include addition to, *alteration* of, replacement or relocation of any standpipe, fire protection water supply, *automatic sprinkler system*, fire alarm system or other work affecting fire protection or life safety.

[C] This section distinguishes between alterations, wherein the code is to be applicable, and ordinary repairs, which are maintenance activities that do not require a permit.

[A] 105.1.6 Annual permit. Instead of an individual construction permit for each *alteration* to an already *approved* system or equipment installation, the *fire code official* is authorized to issue an annual permit on application therefor to any person, firm or corporation regularly employing one or more qualified tradespersons in the building, structure or on the premises owned or operated by the applicant for the permit.

[C] In some instances, such as large buildings or industrial facilities, the repair, replacement or alteration of electrical, gas, mechanical or plumbing systems occurs on a frequent basis. This section allows the fire code official to issue an annual permit for such work. This relieves both the building department and the owners of such facilities from the burden of filing and processing individual applications for this activity; however, there are restrictions on who is entitled to these permits. They can be issued only for work on a previously approved installation and only to an individual or corporation that employs persons specifically qualified in the trade for which the permit is issued. If tradespeople who perform the work involved are required to be licensed in the jurisdiction, then only those persons would be permitted to perform the work. If trade licensing is not required, then the fire code official needs to review and approve the qualifications of the persons who will perform the work. The annual permit can apply only to the individual property that is owned or operated by the applicant.

[A] 105.1.6.1 Annual permit records. The person to whom an annual permit is issued shall keep a detailed record of *alterations* made under such annual permit. The *fire code official* shall have access to such records at all times or such records shall be filed with the *fire code official* as designated.

[C] The work performed in accordance with an annual permit must be inspected by the fire code official, so it is necessary to know the location of such work and when it was performed. This can be accomplished by having records of the work available to the fire code official either at the premises or in the official's office, as determined by the official.

[A] 105.2 Application. Application for a permit required by this code shall be made to the *fire code official* in such form and detail as prescribed by the *fire code official*. Applications for permits shall be accompanied by such plans as prescribed by the *fire code official*.

[C] Applications provided by the jurisdiction should be complete, concise and relevant. Though the burden of proof is on the applicant to supply all necessary information to determine compliance with the code provisions, it is the fire code official's duty to request sufficient information to make a reasonable and informed judgment prior to approving a permit.

[A] 105.2.1 Refusal to issue permit. If the application for a permit describes a use that does not conform to the requirements of this code and other pertinent laws and ordinances, the *fire code official* shall not issue a permit, but shall return the application to the applicant with the refusal to issue such permit. Such refusal shall, where requested, be in writing and shall contain the reasons for refusal.

[C] The fire code official has the option to not issue a permit if the application describes a use that does not conform to the requirements of the code. Note that this direction is not advisory. The fire code official would be in violation of the code if a permit were issued in such circumstances. Similar to Section 104.2.3.2, where requested by the applicant, this section requires the reason for refusal.

[A] 105.2.2 Inspection authorized. Before a new operational permit is *approved*, the *fire code official* is authorized to inspect the receptacles, vehicles, buildings, devices, premises, storage spaces or areas to be used to determine compliance with this code or any operational constraints required.

[C] The inspections described in this section are necessary for the fire code official to determine that the application for an operational permit complies with the code prior to issuing that permit. Operations may not proceed without an operational permit.

[A] 105.2.3 Time limitation of application. An application for a permit for any proposed work or operation shall be deemed to have been abandoned 180 days after the date of filing, unless such application has been diligently prosecuted or a permit shall have been issued; except that the *fire code official* is authorized to grant one or more extensions of time for additional periods not exceeding 90 days each. The extension shall be requested in writing and justifiable cause demonstrated.

[C] Permit applications are not allowed to linger indefinitely in an incomplete condition. Filing for a permit under one edition of the code and then finishing the application process later could create a situation where code requirements have changed, but the permit only requires compliance with the previous code. This section establishes 180 days as the time limit for the permit applicant to provide the fire code official with sufficient information to evaluate any correction com-

ments and take appropriate action. That period should normally be more than enough time for an applicant to satisfy code requirements for submittal of construction documents and all other required information.

There may be circumstances, however, that could cause an application to age beyond 180 days prior to permit issuance, such as awaiting issuance of a report by a quality assurance agency. If the fire code official is satisfied that every effort is being made by the applicant to pursue the application, an extension of time would be acceptable.

[A] 105.2.4 Action on application. The *fire code official* shall examine or cause to be examined applications for permits and amendments thereto within a reasonable time after filing. If the application or the *construction documents* do not conform to the requirements of pertinent laws, the *fire code official* shall reject such application in writing, stating the reasons therefor. If the *fire code official* is satisfied that the proposed work or operation conforms to the requirements of this code and laws and ordinances applicable thereto, the *fire code official* shall issue a permit therefor as soon as practicable.

C This section requires the fire code official to act with reasonable speed on a permit application. In some instances, this time period is set by state or local law. The fire code official must refuse to issue a permit when the application and accompanying documents do not conform to the code. In order to ensure effective communication and due process of law, the reasons for denial of an application for a permit are required to be in writing. Once the fire code official determines that the work described conforms to the code and other applicable laws, the permit must be issued upon payment of the fees required by Section 107.

[A] 105.3 Conditions of a permit. A permit shall constitute permission to maintain, store or handle materials; or to conduct processes that produce conditions hazardous to life or property; or to install equipment utilized in connection with such activities; or to install or modify any *fire protection system* or equipment or any other construction, equipment installation or modification in accordance with the provisions of this code where a permit is required by Section 105.5 or 105.6. Such permission shall not be construed as authority to violate, cancel or set aside any of the provisions of this code or other applicable regulations or laws of the jurisdiction.

C In effect, a permit is a contract or covenant between the jurisdiction and the applicant, allowing the applicant to operate, perform, conduct or direct a hazardous operation, process or occupancy. As with all contracts, the terms remain binding for a finite period. This process allows continual review of the applicant's compliance with the contract's terms. Failure to meet the terms of the contract may result in the applicant's forfeiture of the right to conduct or operate the process, operation or occupancy, and subsequently the fire code official may revoke the permit without further notice.

This section also states the fundamental premise that the permit is only a license to proceed with the work. It is not a license to violate, cancel or set aside any provisions of the code. This is significant because, even if there are errors or oversights in the permit approval process, the permit applicant, not the fire code official, is responsible for code compliance.

[A] 105.3.1 Expiration. An operational permit shall remain in effect until reissued, renewed or revoked, or for such a period of time as specified in the permit. Construction permits shall automatically become invalid unless the work authorized by such permit is commenced within 180 days after its issuance, or if the work authorized by such permit is suspended or abandoned for a period of 180 days after the time the work is commenced. Before such work recommences, a new permit shall be first obtained and the fee to recommence work, if any, shall be one-half the amount required for a new permit for such work, provided that changes have not been made and will not be made in the original *construction documents* for such work, and provided further that such suspension or abandonment has not exceeded one year. Permits are not transferable and any change in occupancy, operation, tenancy or ownership shall require that a new permit be issued.

C A construction permit is invalid when 180 days go by without any of the authorized work being done. The permit holder should be notified in writing that the permit is invalid, including the reasons why.

Permits are neither transferable nor assignable because they are agreements between two specific parties: the fire code official, who is acting for the jurisdiction, and the applicant. Any changes amending the application or terms of the original agreement will require a new application and permit approval.

[A] 105.3.2 Extensions. A permittee holding an unexpired permit shall have the right to apply for an extension of the time within which the permittee will commence work under that permit where work is unable to be commenced within the time required by this section for good and satisfactory reasons. The *fire code official* is authorized to grant, in writing, one or more extensions of the time period of a permit for periods of not more than 180 days each. Such extensions shall be requested by the permit holder in writing and justifiable cause demonstrated.

C The significant issue in this section is that an extension of time is to be granted when justifiable cause is demonstrated by the permit applicant. For example, a construction permit might be granted for certain equipment installation, but the equipment might not be received at the site until after the installation permit expires. To get a time extension, the applicant is to submit a request in writing to the fire code official, including a written explanation of why the work did not proceed within the permit time frame. Although it is customary for a project to begin immediately following issuance of a permit, there may be occasions when an unforeseen delay may occur. This section affords the permit holder an opportunity to apply for and receive a single 180-day extension of time within which to begin a project under a still-valid permit

(less than 180 days old). The applicant must, however, give the fire code official an adequate explanation for the delay in starting a project, which could include such things as the need to obtain approvals or permits for the project from other agencies having jurisdiction. This section requires the fire code official to determine what constitutes “good and satisfactory” reasons for any delay.

[A] 105.3.3 Occupancy prohibited before approval. The building or structure shall not be occupied prior to the *fire code official* issuing a permit and conducting associated inspections indicating the applicable provisions of this code have been met.

[C] The owner of an existing structure may request that the fire code official issue a certificate of occupancy for the structure, provided that there are no pending violations. A final inspection is usually done to verify that the work covered by the permit has been completed in accordance with the code.

[A] 105.3.4 Conditional permits. Where permits are required and on the request of a permit applicant, the *fire code official* is authorized to issue a conditional permit to occupy the premises or portion thereof before the entire work or operations on the premises is completed, provided that such portion or portions will be occupied safely prior to full completion or installation of equipment and operations without endangering life or public welfare. The *fire code official* shall notify the permit applicant in writing of any limitations or restrictions necessary to keep the permit area safe. The holder of a conditional permit shall proceed only to the point for which approval has been given, at the permit holder’s own risk and without assurance that approval for the occupancy or the utilization of the entire premises, equipment or operations will be granted.

[C] The fire code official is allowed to issue a conditional permit prior to the completion of all work. Such a permit is to be issued only when the building or structure is available for safe occupancy prior to full completion. The permit is intended to acknowledge that some building features may not be completed even though the building is safe for occupancy.

[A] 105.3.5 Posting the permit. Issued permits shall be kept on the premises designated therein at all times and shall be readily available for inspection by the *fire code official*.

[C] The permit is not required to be posted, but it is to be kept on-site at all times for inspection by the fire code official.

[A] 105.3.6 Compliance with code. The issuance or granting of a permit shall not be construed to be a permit for, or an approval of, any violation of any of the provisions of this code or of any other ordinance of the jurisdiction. Permits presuming to give authority to violate or cancel the provisions of this code or other ordinances of the jurisdiction shall not be valid. The issuance of a permit based on *construction documents* and other data shall not prevent the *fire code official* from requiring the correction of errors in the *construction documents* and other data. Any addition to or alteration of *approved construction documents* shall be *approved* in advance by the *fire code official*, as evidenced by the issuance of a new or amended permit.

[C] This section includes an important principle regarding construction documents. The fire code official has the authority to require that errors in construction be corrected, even if the construction is based on documents that were part of the applicant’s submittal for a construction permit. Thus, the code requirements are not set aside by approved drawings that may include noncomplying items of construction. Any changes amending the application or construction of the original agreement will require a new application and permit approval.

[A] 105.3.7 Information on the permit. The *fire code official* shall issue all permits required by this code on an *approved* form furnished for that purpose. The permit shall contain a general description of the operation or occupancy and its location and any other information required by the *fire code official*. Issued permits shall bear the signature of the *fire code official* or other *approved* legal authorization.

[C] This section describes the form of the permit and requires that it be either signed by the fire code official or otherwise reflect the legal authorization of the jurisdiction. In many jurisdictions, permits are electronically generated and do not require a traditional signature.

[A] 105.3.8 Validity of permit. The issuance or granting of a permit shall not be construed to be a permit for, or an approval of, any violation of any of the provisions of this code or of any other ordinances of the jurisdiction. Permits presuming to give authority to violate or cancel the provisions of this code or other ordinances of the jurisdiction shall not be valid. The issuance of a permit based on *construction documents*, operational documents and other data shall not prevent the *fire code official* from requiring correction of errors in the documents or other data.

[C] This section states the fundamental premise that the permit is only a license to proceed with the work. It is not a license to violate, cancel or set aside any provisions of the code. This is significant because it means that despite any errors or oversights in the approval process, the permit applicant, not the fire code official, is responsible for code compliance. Also, the permit can be revoked in accordance with Section 105.4.

[A] 105.4 Revocation. The *fire code official* is authorized to revoke a permit issued under the provisions of this code where it is found by inspection or otherwise that there has been a false statement or misrepresentation as to the material facts in the application or *construction documents* on which the permit or approval was based including, but not limited to, any one of the following:

1. The permit is used for a location or establishment other than that for which it was issued.
2. The permit is used for a condition or activity other than that listed in the permit.
3. Conditions and limitations set forth in the permit have been violated.

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4. There have been any false statements or misrepresentations as to the material fact in the application for permit or plans submitted or a condition of the permit.
5. The permit is used by a different person or firm than the name for which it was issued.
6. The permittee failed, refused or neglected to comply with orders or notices duly served in accordance with the provisions of this code within the time provided therein.
7. The permit was issued in error or in violation of an ordinance, regulation or this code.

C The fire code official must revoke all permits shown to be based, all or in part, on any false statement or misinterpretation of fact. An applicant may subsequently reapply for a permit. The code specifies seven specific conditions that allow the fire code official to revoke a permit.

105.5 Required operational permits. The *fire code official* is authorized to issue operational permits for the operations set forth in Sections 105.5.2 through 105.5.53.

C Sections 105.5.1 through 105.5.57 list the conditions requiring operational permits. Many of the items are stated in general terms, in which case the fire code official is to determine whether a specific operation is a significant hazard that requires a permit.

105.5.1 Additive manufacturing. An operational permit is required to conduct *additive manufacturing* operations regulated by Section 4106.3.

C See Section 320 for additive manufacturing requirements.

105.5.2 Aerosol products, aerosol cooking spray products and plastic aerosol 3 products. An operational permit is required to manufacture, store or handle an aggregate quantity of Level 2 or Level 3 aerosol products, aerosol cooking spray products or plastic aerosol 3 products in excess of 500 pounds (227 kg) net weight.

C See Chapter 51 for code requirements covering aerosol products (see commentary, Section 105.5).

105.5.3 Amusement areas. An operational permit is required to operate a *special amusement area*.

C For requirements that apply to special amusement areas, see Sections 202, 907.2.12 and 914.7, and IBC Section 411 (see commentary, Section 105.5).

105.5.4 Aviation facilities. An operational permit is required to use a Group H or Group S occupancy for aircraft servicing or repair and aircraft fuel-servicing vehicles. Additional permits required by other sections of this code include, but are not limited to, hot work, hazardous materials and flammable or combustible finishes.

C See Chapter 20 for aviation facility requirements (see commentary, Section 105.5).

105.5.5 Carnivals and fairs. An operational permit is required to conduct a carnival or fair.

C See Section 3103.3 for carnival requirements (see commentary, Section 105.5).

105.5.6 Cellulose nitrate film. An operational permit is required to store, handle or use cellulose nitrate film in a Group A occupancy.

C Although cellulose nitrate film is no longer in general use, there are a small number of locations in which this type of film is archived or restored for historical purposes. This section applies to those few locations (see Section 306 for cellulose nitrate film requirements).

105.5.7 Combustible dust-producing operations. An operational permit is required to operate a grain elevator, flour starch mill, feed mill, or a plant pulverizing aluminum, coal, cocoa, magnesium, spices or sugar, or other operations producing *combustible dusts* as defined in Chapter 2.

C See Chapter 22 for combustible dust-producing operations (see commentary, Section 105.5).

105.5.8 Combustible fibers. An operational permit is required for the storage and handling of *combustible fibers* in quantities greater than 100 cubic feet (2.8 m³).

Exception: A permit is not required for agricultural storage.

C See Chapter 37 for combustible fiber requirements. The exception is for agricultural storage facilities where the hazard to persons is minimal (see Section 105.5).

105.5.9 Compressed gases. An operational permit is required for the storage, use or handling at *normal temperature and pressure* (NTP) of *compressed gases* in excess of the amounts listed in Table 105.5.9.

Exception: Vehicles equipped for and using *compressed gas* as a fuel for propelling the vehicle.

C See Chapter 53 for compressed gas requirements. The exception exempts vehicles equipped for compressed gas, since the code requirements for compressed gases do not apply to them.

TABLE 105.5.9—PERMIT AMOUNTS FOR COMPRESSED GASES

TYPE OF GAS	AMOUNT (cubic feet at NTP)
Carbon dioxide used in carbon dioxide enrichment systems	875 (100 lb)
Carbon dioxide used in insulated liquid carbon dioxide beverage dispensing applications	875 (100 lb)
Corrosive	200
Flammable (except cryogenic fluids and liquefied petroleum gases)	200
Highly toxic	Any Amount
Inert and simple asphyxiant	6,000
Oxidizing (including oxygen)	504
Pyrophoric	Any Amount
Toxic	Any Amount
For SI: 1 cubic foot = 0.02832 m ³ .	

C Where the use of compressed gases exceeds the amounts indicated in Table 105.5.9, an operational permit is required. The quantities in the table are at normal temperature and pressure (NTP) (see Chapter 53 for compressed gas requirements).

105.5.10 Covered and open mall buildings. An operational permit is required for:

1. The placement of retail fixtures and displays, concession equipment, displays of highly combustible goods and similar items in the mall.
2. The display of liquid- or gas-fired equipment in the mall.
3. The use of open-flame or flame-producing equipment in the mall.

C The listed operations in a covered or open mall building require an operational permit since they involve a significant hazard to the occupants. See Section 308 for open-flame regulations (see commentary, Section 105.5).

105.5.11 Cryogenic fluids. An operational permit is required to produce, store, transport on site, use, handle or dispense *cryogenic fluids* in excess of the amounts listed in Table 105.5.11.

Exception: Permits are not required for vehicles equipped for and using *cryogenic fluids* as a fuel for propelling the vehicle or for refrigerating the lading.

C See Chapter 55 for requirements regarding cryogenic fluids. The exception exempts vehicles using cryogenic fluids since the code requirements do not apply to them.

TABLE 105.5.11—PERMIT AMOUNTS FOR CRYOGENIC FLUIDS

TYPE OF CRYOGENIC FLUID	INSIDE BUILDING (gallons)	OUTSIDE BUILDING (gallons)
Flammable	More than 1	60
Inert	60	500
Oxidizing (includes oxygen)	10	50
Physical or health hazard not indicated above	Any Amount	Any Amount
For SI: 1 gallon = 3.785 L.		

C Where cryogenic fluids are used in excess of the amounts shown in Table 105.5.11, an operational permit is required. The listed amounts are significantly different inside or outside of a building since the hazard is greatly reduced if a leak occurs outdoors.

105.5.12 Cutting and welding. An operational permit is required to conduct cutting or welding operations within the jurisdiction.

C See Chapter 35 for welding requirements (see commentary, Section 105.5).

105.5.13 Dry cleaning. An operational permit is required to engage in the business of dry cleaning or to change to a more hazardous cleaning solvent used in existing dry cleaning equipment.

C See Chapter 21 for dry cleaning regulations (see commentary, Section 105.5).

105.5.14 Energy storage systems. An operational permit is required for stationary and mobile energy storage systems regulated by Section 1207.

C See Section 1207 for energy storage system requirements.

105.5.15 Exhibits and trade shows. An operational permit is required to operate exhibits and trade shows.

C The primary concern is to identify hazardous and highly flammable materials that could be involved in an exhibit or booth (see commentary, Section 105.5).

105.5.16 Explosives. An operational permit is required for the manufacture, storage, handling, sale or use of any quantity of *explosives, explosive materials*, fireworks or pyrotechnic special effects within the scope of Chapter 56.

Exception: Storage in Group R-3 occupancies of smokeless propellant, black powder and small arms primers for personal use, not for resale and in accordance with Section 5606.

C See Chapter 56 for requirements for explosives and fireworks (see commentary, Section 105.5 and Chapter 56).

The exception correlates the permit requirements for the possession, storage or use of smokeless propellant, black powder and small arms primers for personal use in Group R-3 residential occupancies with the scope of Chapter 56, as stated in Section 5601.1, Exception 4 and Section 5606.4. The exception is also consistent with NFPA 495, referenced in Chapter 56, which limits quantities allowed in residences, but allows for quantities in residences outside the scope of Chapter 56 to be regulated without a permit.

105.5.17 Fire hydrants and valves. An operational permit is required to use or operate fire hydrants or valves intended for fire suppression purposes that are installed on water systems and provided with ready access from a fire apparatus access road that is open to or generally used by the public.

Exception: A permit is not required for authorized employees of the water company that supplies the system or the fire department to use or operate fire hydrants or valves.

C An operational permit is required for persons other than authorized employees of the water company or the fire department to operate fire hydrants or valves. This restriction is intended to make sure that the use will not result in a lack of water supply and pressure that may be needed for firefighting purposes. The exception allows water company employees or the fire department to use fire hydrants or valves without a permit. Such use is common in order to flush out the piping periodically. When fire departments or fire districts interact with water districts, they should communicate the need for the fire department to use the hydrants and valves for nonemergency situations, such as training. A notification procedure is needed to let the water district know of this planned use.

105.5.18 Flammable and combustible liquids. An operational permit is required:

1. To use or operate a pipeline for the transportation within facilities of *flammable* or *combustible liquids*. This requirement shall not apply to the off-site transportation in pipelines regulated by the Department of Transportation (DOTn) nor does it apply to piping systems.
2. To store, handle or use Class I liquids in excess of 5 gallons (19 L) in a building or in excess of 10 gallons (37.9 L) outside of a building, except that a permit is not required for the following:
 - 2.1. The storage or use of Class I liquids in the fuel tank of a motor vehicle, aircraft, motorboat, mobile power plant or mobile heating plant, unless such storage, in the opinion of the *fire code official*, would cause an unsafe condition.
 - 2.2. The storage or use of paints, oils, varnishes or similar flammable mixtures where such liquids are stored for maintenance, painting or similar purposes for a period of not more than 30 days.
3. To store, handle or use Class II or Class IIIA liquids in excess of 25 gallons (95 L) in a building or in excess of 60 gallons (227 L) outside a building, except for fuel oil used in connection with oil-burning equipment.
4. To store, handle or use Class IIIB liquids in tanks or portable tanks for fueling motor vehicles at motor fuel-dispensing facilities or where connected to fuel-burning equipment.

Exception: Fuel oil and used motor oil used for space heating or water heating.
5. To remove Class I or II liquids from an underground storage tank used for fueling motor vehicles by any means other than the *approved*, stationary on-site pumps normally used for dispensing purposes.
6. To operate tank vehicles, equipment, tanks, plants, terminals, wells, fuel-dispensing stations, refineries, distilleries and similar facilities where *flammable* and *combustible liquids* are produced, processed, transported, stored, dispensed or used.
7. To place temporarily out of service (for more than 90 days) an underground, protected above-ground or above-ground *flammable* or *combustible liquid* tank.
8. To change the type of contents stored in a *flammable* or *combustible liquid* tank to a material that poses a greater hazard than that for which the tank was designed and constructed.
9. To manufacture, process, blend or refine *flammable* or *combustible liquids*.
10. To engage in the dispensing of liquid fuels into the fuel tanks of motor vehicles at commercial, industrial, governmental or manufacturing establishments in accordance with Section 5706.5.4 or to engage in on-demand *mobile fueling* operations in accordance with Section 5707.
11. To utilize a site for the dispensing of liquid fuels from tank vehicles into the fuel tanks of motor vehicles, marine craft and other special equipment at commercial, industrial, governmental or manufacturing establishments in accordance with

Section 5706.5.4 or, where required by the *fire code official*, to utilize a site for on-demand *mobile fueling* operations in accordance with Section 5707.

C See Chapter 57 for regulations regarding flammable and combustible liquids (see commentary, Section 105.5).

105.5.19 Floor finishing. An operational permit is required for floor finishing or surfacing operations exceeding 350 square feet (33 m²) using Class I or Class II liquids.

C The concern of this section is the proper use and handling of Class I or II liquids that are used in the floor-finishing process. If such liquids are not used, an operational permit is not required for floor finishing.

105.5.20 Fruit and crop ripening. An operational permit is required to operate a fruit- or crop-ripening facility or conduct a fruit-ripening process using ethylene gas.

C See Chapter 25 for regulations for fruit- and crop-ripening processes where ethylene gas is used (see commentary, Section 105.5).

105.5.21 Fumigation and insecticidal fogging. An operational permit is required to operate a business of fumigation or insecticidal fogging, and to maintain a room, vault or chamber in which a toxic or flammable fumigant is used.

C See Chapter 26 for fumigation and insecticidal fogging regulations within structures (see commentary, Section 105.5).

105.5.22 Hazardous materials. An operational permit is required to store, transport on site, dispense, use or handle hazardous materials in excess of the amounts listed in Table 105.5.22.

C See Chapter 50 for the general provisions regarding hazardous materials. Also see Chapters 51 through 67 for regulations regarding a specific hazardous material (see commentary, Section 105.5).

TABLE 105.5.22—PERMIT AMOUNTS FOR HAZARDOUS MATERIALS

TYPE OF MATERIAL	AMOUNT
Combustible liquids	See Section 105.5.18
Corrosive materials	
Gases	See Section 105.5.9
Liquids	55 gallons
Solids	1,000 pounds
Explosive materials	See Section 105.5.16
Flammable materials	
Gases	See Section 105.5.9
Liquids	See Section 105.5.18
Solids	100 pounds
Highly toxic materials	
Gases	See Section 105.5.9
Liquids	Any Amount
Solids	Any Amount
Organic peroxides	
Liquids	
Class I	Any Amount
Class II	Any Amount
Class III	1 gallon
Class IV	2 gallons
Class V	No Permit Required
Solids	
Class I	Any Amount
Class II	Any Amount
Class III	10 pounds
Class IV	20 pounds
Class V	No Permit Required
Oxidizing materials	
Gases	See Section 105.5.9

TABLE 105.5.22—PERMIT AMOUNTS FOR HAZARDOUS MATERIALS—continued	
TYPE OF MATERIAL	AMOUNT
Liquids	
Class 4	Any Amount
Class 3	1 gallon ^a
Class 2	10 gallons
Class 1	55 gallons
Solids	
Class 4	Any Amount
Class 3	10 pounds ^b
Class 2	100 pounds
Class 1	500 pounds
Pyrophoric materials	
Gases	Any Amount
Liquids	Any Amount
Solids	Any Amount
Toxic materials	
Gases	See Section 105.5.9
Liquids	10 gallons
Solids	100 pounds
Unstable (reactive) materials	
Liquids	
Class 4	Any Amount
Class 3	Any Amount
Class 2	5 gallons
Class 1	10 gallons
Solids	
Class 4	Any Amount
Class 3	Any Amount
Class 2	50 pounds
Class 1	100 pounds
Water-reactive materials	
Liquids	
Class 3	Any Amount
Class 2	5 gallons
Class 1	55 gallons
Solids	
Class 3	Any Amount
Class 2	50 pounds
Class 1	500 pounds
For SI: 1 gallon = 3.785 L, 1 pound = 0.454 kg.	
a. 22 gallons where Table 5003.1.1(1) Note j applies and hazard identification signs in accordance with Section 5003.5 are provided for quantities of 22 gallons or less.	
b. 220 pounds where Table 5003.1.1(1) Note j applies and hazard identification signs in accordance with Section 5003.5 are provided for quantities of 220 pounds or less.	

C Where the amounts of hazardous materials in the table are exceeded, an operational permit is required. This applies to the storage, transportation on site, dispensing, use or handling of the hazardous materials that are listed in the table. Notes a and b create parity for Class 3 oxidizer permit amounts with the adjustments to the maximum allowable quantity (MAQ) per control area specified in Table 5003.1.1(1), Note j. This eliminates the need for small apartment complexes and similar occupancies with swimming pools to obtain permits for normal pool maintenance using relatively small amounts of material.

105.5.23 HPM facilities. An operational permit is required to store, handle or use hazardous production materials.

C See Chapter 27 for regulations regarding semiconductor fabrication facilities (see commentary, Section 105.5).

105.5.24 High-piled storage. An operational permit is required to use a building or portion thereof with more than 500 square feet (46 m²), including aisles, of *high-piled combustible storage*.

C See Chapter 32 for high-piled storage provisions (see commentary, Section 105.5).

105.5.25 Hot work operations. An operational permit is required for hot work including, but not limited to:

1. Public exhibitions and demonstrations where hot work is conducted.
2. Use of portable hot work equipment inside a structure.

Exception: Work that is conducted under a construction permit.

3. Fixed-site hot work equipment, such as welding booths.
4. Hot work conducted within a wildfire risk area.
5. Application of roof coverings with the use of an open-flame device.
6. Where *approved*, the *fire code official* shall issue a permit to carry out a hot work program. This program allows *approved* personnel to regulate their facility's hot work operations. The *approved* personnel shall be trained in the fire safety aspects denoted in this chapter and shall be responsible for issuing permits requiring compliance with the requirements found in Chapter 35. These permits shall be issued only to their employees or hot work operations under their supervision.

C See Chapter 35 for hot work regulations. The exception to Item 2 in this section recognizes that work done under a construction permit is already covered by that permit, so an operational permit is not required (see commentary, Section 105.5).

105.5.26 Indoor plant cultivation. An operational permit is required for plant cultivation where a carbon dioxide (CO₂) enriched environment is created.

C See Chapter 39 for regulations regarding indoor plant cultivation (see commentary, Section 105.5).

105.5.27 Industrial ovens. An operational permit is required for operation of industrial ovens regulated by Chapter 30.

C See Chapter 30 for regulations regarding industrial ovens (see commentary, Section 105.5).

105.5.28 Liquid- or gas-fueled vehicles or equipment in assembly buildings. An operational permit is required to display, operate or demonstrate liquid- or gas-fueled vehicles or equipment in assembly buildings.

C See Section 314.4 for requirements regarding liquid- or gas-fueled vehicles inside buildings (see commentary, Section 105.5).

105.5.29 Lithium batteries. An operational permit is required for an accumulation of more than 15 cubic feet (0.42 m³) of lithium-ion and lithium metal batteries, where required by Section 320.2.

C See Section 320.1 for requirements regarding the storage of lithium-ion batteries (see commentary, Section 105.5).

105.5.30 LP-gas. An operational permit is required for:

1. Storage and use of LP-gas.

Exception: A permit is not required for individual containers with a 500-gallon (1893 L) water capacity or less or multiple container systems having an aggregate quantity not exceeding 500 gallons (1893 L), serving occupancies in Group R-3.

2. Operation of cargo tankers that transport LP-gas.

C See Chapter 61 for liquefied petroleum gas (LP-gas) regulations. The exception to Item 1 in this section exempts small tanks with an individual capacity of 500 gallons (1893 L) or multiple small tanks with an aggregate capacity of 500 gallons (1893 L) commonly found in residential service. A permit is required where the aggregate quantity of multiple small LP-gas containers exceeds 500 gallons (1893 L). It has become commonplace for LP-gas distributors to install LP-gas systems exceeding 500 gallons (1893 L) that consist of multiple containers in series with individual containers that do not exceed 500 gallons (1893 L), thereby avoiding the permit requirement. It is appropriate to require a permit at these locations given the significant hazard associated with these quantities. Item 2 covers cargo tankers, since they transport LP-gas onto premises covered by the code and, therefore, represent a potential hazard.

105.5.31 Lumber yards and woodworking plants. An operational permit is required for the storage or processing of lumber exceeding 100,000 board feet (8,333 ft³) (236 m³).

C See Chapter 28 for provisions for lumber yards and woodworking plants (see commentary, Section 105.5).

105.5.32 Magnesium. An operational permit is required to melt, cast, heat treat or grind more than 10 pounds (4.54 kg) of magnesium.

C See Section 5906 for the code requirements for magnesium (see commentary, Section 105.5).

105.5.33 Miscellaneous combustible storage. An operational permit is required to store in any building or on any premises in excess of 2,500 cubic feet (71 m³) gross volume of combustible empty packing cases, boxes, barrels or similar containers, combustible pallets, rubber tires, rubber, cork or similar combustible material.

C See Section 315 for requirements for miscellaneous combustible material storage (see commentary, Section 105.5).

105.5.34 Mobile food preparation vehicles. An operational permit is required for mobile food preparation vehicles equipped with appliances that produce smoke or grease-laden vapors.

C See Section 4106 for requirements for mobile food preparation vehicles (see commentary, Section 105.5).

105.5.35 Motor fuel-dispensing facilities. An operational permit is required for the operation of automotive, marine and fleet motor fuel-dispensing facilities.

C See Chapter 23 for requirements for motor fuel-dispensing facilities (see commentary, Section 105.5).

105.5.36 Open burning. An operational permit is required for the kindling or maintaining of an open fire or a fire on any public street, alley, road, or other public or private ground. Instructions and stipulations of the permit shall be complied with.

Exception: *Recreational fires.*

C See Section 307 for open burning provisions. Section 202 includes the definition of “Open burning.” The exception exempts recreational fires, which are also defined in Section 202.

105.5.37 Open flames and torches. An operational permit is required to remove paint with a torch, or to use a torch or open-flame device in a wildfire risk area.

C See Section 308 for regulations regarding open flames (see commentary, Section 105.5).

105.5.38 Open flames and candles. An operational permit is required to use open flames or candles in connection with assembly areas, dining areas of restaurants or drinking establishments.

C See Section 308 for regulations regarding open flames (see commentary, Section 105.5).

105.5.39 Organic coatings. An operational permit is required for any organic-coating manufacturing operation producing more than 1 gallon (4 L) of an organic coating in one day.

C The manufacture of organic coatings is addressed in Chapter 29 (see commentary, Section 105.5).

105.5.40 Outdoor assembly event. An operational permit is required to conduct an *outdoor assembly event* where planned attendance exceeds 1,000 persons.

C See Section 3107 for requirements regarding outdoor assembly events (see commentary, Section 105.5).

105.5.41 Places of assembly. An operational permit is required to operate a place of assembly.

C Because of the higher occupant loads found in Group A occupancies, such occupancies justify an increased level of scrutiny, such as is provided through the permit process.

105.5.42 Plant extraction systems. An operational permit is required to use plant extraction systems.

C See Chapter 39 for regulations regarding plant extraction systems. More specifically, this chapter addresses the processing and extraction of oils from plants (see commentary, Section 105.5).

105.5.43 Private fire hydrants. An operational permit is required for the removal from service, use or operation of private fire hydrants.

Exception: A permit is not required for private industry with trained maintenance personnel, private fire brigade or fire departments to maintain, test and use private hydrants.

C The purpose of an operational permit for the removal of private fire hydrants is to see that adequate fire hydrants are maintained for use during a fire. The exception allows testing and use of private fire hydrants by trained private industry personnel without an operational permit.

105.5.44 Pyrotechnic special effects material. An operational permit is required for use and handling of pyrotechnic special effects material.

C See Chapter 56 for fireworks regulations. The definition of “Pyrotechnic special-effect material” is found in Section 202 (see commentary, Section 105.5).

105.5.45 Pyroxylin plastics. An operational permit is required for storage or handling of more than 25 pounds (11 kg) of cellulose nitrate (pyroxylin) plastics, and for the assembly or manufacture of articles involving pyroxylin plastics.

C See Chapter 65 for requirements regarding pyroxylin (cellulose nitrate) plastics (see commentary, Section 105.5).

105.5.46 Refrigeration equipment. An operational permit is required to operate a mechanical refrigeration unit or system regulated by Chapter 6.

C See Section 608 for mechanical refrigeration regulations (see commentary, Section 105.5).

105.5.47 Repair garages and motor fuel-dispensing facilities. An operational permit is required for operation of repair garages.

C See Chapter 23 for requirements for repair garages (see commentary, Section 105.5).

105.5.48 Rooftop heliports. An operational permit is required for the operation of a rooftop heliport.

C See Chapter 20 for aviation facility requirements. Section 2007 contains helistop and heliport requirements (see commentary, Section 105.5).

105.5.49 Spraying or dipping. An operational permit is required to conduct a spraying or dipping operation utilizing *flammable* or *combustible liquids*, or the application of combustible powders regulated by Chapter 24.

C See Chapter 24 for flammable finish requirements. Section 2404 contains the spray finishing provisions, Section 2405 addresses dipping operations and Section 2406 includes powder coating regulations (see commentary, Section 105.5).

105.5.50 Storage of scrap tires and tire byproducts. An operational permit is required to establish, conduct or maintain storage of scrap tires and tire byproducts that exceeds 2,500 cubic feet (71 m³) of total volume of scrap tires, and for indoor storage of tires and tire byproducts.

C See Chapter 34 for regulations regarding tire rebuilding and tire storage (see Section 105.5).

105.5.51 Temporary membrane structures, special event structures and tents. An operational permit is required to operate an air-supported temporary membrane structure, a temporary *special event structure* or a tent having an area in excess of 400 square feet (37 m²).

Exceptions:

1. Tents used exclusively for recreational camping purposes.
2. Tents, curtains and extensions attached thereto, when used for funeral services.
3. Tents open on all sides, which comply with all of the following:
 - 3.1. Individual tents having a maximum size of 700 square feet (65 m²).
 - 3.2. The aggregate area of multiple tents placed side by side without a fire break clearance of not less than 12 feet (3658 mm) shall not exceed 700 square feet (65 m²) total.
 - 3.3. A minimum clearance of 12 feet (3658 mm) to structures and other tents shall be provided.

C See Chapter 31 for requirements for temporary tents, membrane structures and special event structures. The first exception in this section exempts recreational camping tents, since they are small, temporary and have few occupants. The second exception includes funeral tents where the hazard during typical assembly use is very low. The third exception exempts relatively small tents that are very low hazard, since they are spaced at least 12 feet (3658 mm) apart (see commentary, Section 3103.2).

105.5.52 Tire-rebuilding plants. An operational permit is required for the operation and maintenance of a tire-rebuilding plant.

C See Chapter 34 for regulations regarding tire rebuilding and tire storage (see Section 105.5).

105.5.53 Waste handling. An operational permit is required for the operation of wrecking yards, junk yards and waste material-handling facilities.

C See Section 315 for miscellaneous combustible materials storage requirements, Section 5004 for provisions regarding the storage of hazardous materials and Section 2808 for provisions regarding yard waste and recycling facilities (see commentary, Section 105.5).

105.5.54 Wood products. An operational permit is required to store chips, hogged material, lumber or plywood in excess of 200 cubic feet (6 m³).

C See Section 2808 for requirements regarding the storage and handling of wood chips, hogged material, fines, compost and raw product in association with yard waste and recycling facilities (see commentary, Section 105.5).

105.5.55 Temporary heating or cooking in tents or membrane structures. An operational permit is required to operate temporary heating or cooking equipment within tents or membrane structures.

C See Chapter 41 for regulations regarding temporary heating or cooking in tents or membrane structures (see Section 105.5).

105.5.56 Temporary heating or cooking in wildfire risk areas. Where required by local regulations, an operational permit is required to operate temporary heating or cooking equipment in wildfire risk areas.

C See Chapter 41 for regulations regarding temporary heating or cooking in wildfire risk areas (see Section 105.5).

105.5.57 Temporary heating for construction sites. An operational permit is required to operate temporary heating equipment in structures during the course of construction, *alteration* or demolition.

C See Chapter 33 for regulations regarding temporary heating at construction sites (see Section 105.5).

[A] 105.6 Required construction permits. The *fire code official* is authorized to issue construction permits for work as set forth in Sections 105.6.1 through 105.6.25.

C This section addresses conditions requiring a construction permit. Generally, a construction permit is required when a safety-related system or hazardous material storage is installed or an existing system or facility is modified. Other sections of the code may also apply.

In some cases, the requirements in Sections 105.6.1 through 105.6.25 are stated in only general terms. In these instances, the fire code official is to evaluate the scope of work involved for the modification or installation and determine whether a construction permit is required for the specific project.

[A] 105.6.1 Automatic fire-extinguishing systems. A construction permit is required for installation of or modification to an *automatic fire-extinguishing system*, other than an *automatic sprinkler system*. Maintenance performed in accordance with this code is not considered to be a modification and does not require a permit.

C See Chapter 9 for fire protection system requirements. A construction permit is required for the installation or modification of an automatic fire-extinguishing system so that the work can be verified to meet the code requirements, since the system is obviously safety related (see commentary, Section 105.6).

[A] 105.6.2 Automatic sprinkler systems. A construction permit is required for installation of or modification to an *automatic sprinkler system*. Maintenance performed in accordance with this code is not considered to be a modification and does not require a permit.

C See Chapter 9 for fire protection system requirements. A construction permit is required for the installation or modification of an automatic fire sprinkler system so that the work can be verified to meet the code requirements, since the system is obviously safety related (see commentary, Section 105.6).

[A] 105.6.3 Compressed gases. Where the *compressed gases* in use or storage exceed the amounts listed in Table 105.5.9, a construction permit is required to install, repair damage to, abandon, remove, place temporarily out of service, or close or substantially modify a *compressed gas system*.

Exceptions:

1. Routine maintenance.
2. For emergency repair work performed on an emergency basis, application for permit shall be made within two working days of commencement of work.

C See Chapter 53 for the requirements for compressed gas systems. Where the volume of the compressed gas presents a significant health hazard and the quantity exceeds the allowed amounts in Table 105.5.9, a permit is needed to trigger construction document submittal, document review and inspections of the work on the system. The exceptions exempt maintenance work and allow emergency work to proceed immediately.

[A] 105.6.4 Cryogenic fluids. A construction permit is required for installation of or *alteration* to outdoor stationary *cryogenic fluid* storage systems where the system capacity exceeds the amounts listed in Table 105.5.11. Maintenance performed in accordance with this code is not considered to be an *alteration* and does not require a construction permit.

C See Chapter 55 for the requirements for cryogenic fluids and Chapter 58 for the requirements for flammable cryogenic fluids. The application for a construction permit for cryogenic fluids is intended to trigger a plan review that will examine constraints on location and the requirements of Chapters 55 and 58.

[A] 105.6.5 Emergency responder communication coverage system. A construction permit is required for installation of or modification to in-building, two-way emergency responder communication coverage systems and related equipment. Maintenance performed in accordance with this code is not considered to be a modification and does not require a construction permit.

C This section establishes that a permit must be obtained for the installation or modification of the emergency responder radio coverage system (ERRCS) to ensure that the work is done correctly and that any parts replacement will be compatible with the existing system components. Since the normal maintenance of a system typically would not involve alteration of the system, it would not require a permit.

[A] 105.6.6 Energy storage systems. A construction permit is required to install energy storage systems regulated by Section 1207.

C See Section 1207 for energy storage system requirements. A construction permit is required for the installation of energy storage systems in order to provide a complete and detailed review of these specialized systems to confirm that they meet specific code requirements prior to the start of installation.

[A] 105.6.7 Fire alarm and detection systems and related equipment. A construction permit is required for installation of or modification to fire alarm and detection systems and related equipment. Maintenance performed in accordance with this code is not considered to be a modification and does not require a construction permit.

C See Section 907 for fire alarm and detection requirements. A construction permit is required for installation or modification of these systems since they are obviously safety related. A permit is not required for maintenance when no modifications are made to the systems (see commentary, Section 105.6).

[A] 105.6.8 Fire pumps and related equipment. A construction permit is required for installation of or modification to fire pumps and related fuel tanks, jockey pumps, controllers and generators. Maintenance performed in accordance with this code is not considered to be a modification and does not require a construction permit.

C See Section 913 for requirements regarding fire pumps. A construction permit is required for modification or installation of equipment that is necessary to serve the sprinkler or standpipe system. This construction work must be monitored since these are safety-related systems (see commentary, Section 105.6).

[A] 105.6.9 Flammable and combustible liquids. A construction permit is required:

1. To install, repair or modify a pipeline for the transportation of *flammable* or *combustible liquids*.
2. To install, construct or alter tank vehicles, equipment, tanks, plants, terminals, wells, fuel-dispensing stations, refineries, distilleries and similar facilities where *flammable* and *combustible liquids* are produced, processed, transported, stored, dispensed or used.
3. To install, alter, remove, abandon or otherwise dispose of a *flammable* or *combustible liquid tank*.

[C] See Chapter 57 for provisions for flammable and combustible liquids. This section requires a construction permit for any of the three activities listed, since flammable and combustible liquids are a significant hazard (see commentary, Section 105.6).

[A] 105.6.10 Fuel cell power systems. A construction permit is required to install *stationary fuel cell power systems*.

[C] See Section 1205 for requirements regarding fuel cell power systems (see commentary, Section 105.6).

[A] 105.6.11 Gas detection systems. A construction permit is required for the installation of or modification to *gas detection systems*. Maintenance performed in accordance with this code is not considered a modification and shall not require a permit.

[C] Gas detection systems are required throughout the IFC and are addressed as a life safety system in Section 916 such as for CO₂ systems and toxic and highly toxic gases. See Section 916 for general requirements for gas detection systems (see commentary, Section 105.6).

[A] 105.6.12 Gates and barricades across fire apparatus access roads. A construction permit is required for the installation of or modification to a gate or barricade across a fire apparatus access road.

[C] The permit gives the fire code official the ability to ensure that access requirements are met.

Current code requirements for gates and barricades include an approved method of locking/securing, proper dimensions and opening width, and proper devices for operation. See the commentaries to Sections 503.5 and 503.6.

[A] 105.6.13 Hazardous materials. A construction permit is required to install, repair damage to, abandon, remove, place temporarily out of service, or close or substantially modify a storage facility or other area regulated by Chapter 50 where the hazardous materials in use or storage exceed the amounts listed in Table 105.5.22.

Exceptions:

1. Routine maintenance.
2. For repair work performed on an emergency basis, application for permit shall be made within two working days of commencement of work.

[C] A construction permit is needed for hazardous-material-related construction to ensure submittal of construction documents, document review and inspection of the work for code compliance. The exceptions provide exemptions for maintenance work and allow emergency work to proceed immediately, provided the permit application is submitted within two working days of starting the job.

[A] 105.6.14 High-piled combustible storage. A construction permit is required for the installation of or modification to a structure with more than 500 square feet (46 m²), including aisles, of *high-piled combustible storage*. Maintenance performed in accordance with this code is not considered to be a modification and does not require a construction permit.

[C] See Chapter 32 and Section 3201.3 regarding high-piled combustible storage and the requirements for a construction permit (see commentary, Section 105.6).

[A] 105.6.15 Industrial ovens. A construction permit is required for installation of industrial ovens regulated by Chapter 30.

Exceptions:

1. Routine maintenance.
2. For repair work performed on an emergency basis, application for permit shall be made within two working days of commencement of work.

[C] A construction permit is required for industrial oven installation so that the requirements in Chapter 30 for industrial ovens can be verified. The exceptions provide exemptions for maintenance work and allow emergency work to proceed immediately, provided that the permit is applied for within two working days after work begins.

[A] 105.6.16 LP-gas. A construction permit is required for installation of or modification to an LP-gas system. Maintenance performed in accordance with this code is not considered to be a modification and does not require a permit.

[C] See Chapter 61 for the requirements for LP-gas systems (see commentary, Section 105.6). Ordinary maintenance that does not involve modifications to the system does not require a construction permit.

[A] 105.6.17 Motor vehicle repair rooms and booths. A construction permit is required to install or modify a motor vehicle repair room or booth. Maintenance performed in accordance with this code is not considered to be a modification and does not require a permit.

[C] See Sections 2311.8.3 and 2311.8.4 for the requirements for motor vehicle repair rooms and booths, respectively (see commentary, Section 105.6).

[A] 105.6.18 Plant extraction systems. A construction permit is required for installation of or modification to plant extraction systems. Maintenance performed in accordance with this code is not considered to be a modification and does not require a permit.

C See Chapter 39 for requirements for plant extraction systems. Note that this chapter deals specifically with the processing and extraction of oils from plants. (see commentary, Section 105.6).

[A] 105.6.19 Private fire hydrants. A construction permit is required for the installation or modification of private fire hydrants. Maintenance performed in accordance with this code is not considered to be a modification and does not require a permit.

C A construction permit is needed for the installation or modification of private fire hydrants so that they remain in service for fire protection purposes. The water flow rate and pressure capability must be maintained. Ordinary maintenance that does not involve modifications does not require a construction permit.

[A] 105.6.20 Smoke control or smoke exhaust systems. Construction permits are required for installation of or alteration to smoke control or smoke exhaust systems. Maintenance performed in accordance with this code is not considered to be an *alteration* and does not require a permit.

C Smoke control and smoke exhaust systems are required fire protection systems by Chapter 9. A permit is necessary to ensure proper design and installation. Ordinary maintenance that does not involve modifications to the system does not require a construction permit.

[A] 105.6.21 Solar photovoltaic power systems. A construction permit is required to install or modify solar photovoltaic power systems. Maintenance performed in accordance with this code is not considered to be a modification and does not require a permit.

C Section 1204 regulates solar photovoltaic power system installations on buildings. Because of the unique electrical and physical hazards they present to firefighters and the impact such systems have on fire suppression operations, a permit is required. Ordinary maintenance that does not involve modifications to the system does not require a construction permit.

[A] 105.6.22 Special event structure. A single construction permit is required to erect and take down a *temporary special event structure* in accordance with Section 105.5.51.

C See Section 3105 for specific requirements for temporary special event structures (see commentary, Section 105.6).

[A] 105.6.23 Spraying or dipping. A construction permit is required to install or modify a spray room, dip tank or booth. Maintenance performed in accordance with this code is not considered to be a modification and does not require a permit.

C See Chapter 24 for flammable finish requirements. The spray finishing requirements are in Section 2404, while dipping operations regulations are in Section 2405 (see commentary, Section 105.6). Ordinary maintenance that does not involve modifications does not require a construction permit.

[A] 105.6.24 Standpipe systems. A construction permit is required for the installation, modification or removal from service of a standpipe system. Maintenance performed in accordance with this code is not considered to be a modification and does not require a permit.

C See Section 905 for standpipe system requirements. Construction permits are required for standpipe systems because they are safety-related fire protection systems. Ordinary maintenance that does not involve modifications to the system does not require a construction permit.

[A] 105.6.25 Temporary membrane structures and tents. A construction permit is required to erect an air-supported temporary membrane structure, a temporary special event structure or a tent in accordance with Section 105.5.51.

C See Chapter 31 for requirements regarding tents and other membrane structures.

SECTION 106—CONSTRUCTION DOCUMENTS

[A] 106.1 Submittals. *Construction documents* and supporting data shall be submitted in two or more sets with each application for a permit and in such form and detail as required by the *fire code official*. The *construction documents* shall be prepared by a *registered design professional* where required by the statutes of the jurisdiction in which the project is to be constructed.

Exception: The *fire code official* is authorized to waive the submission of *construction documents* and supporting data not required to be prepared by a *registered design professional* if it is found that the nature of the work applied for is such that review of *construction documents* is not necessary to obtain compliance with this code.

C A detailed description of the work for which an application is made must be submitted in the form and detail required by the fire code official. Construction documents are to be prepared by a registered design professional where required by state laws in effect in the jurisdiction. States have professional registration laws that specify the type of construction documents to be prepared by a registered design professional. The code relies on these state laws to determine where a registered design professional is required.

The requirement for the preparation of construction documents and the submittal of calculations is specified by the code in several chapters. For example, Section 901.2 specifies that construction documents and calculations are to be submitted for fire protection systems when required by the fire code official.

[A] 106.2 Examination of documents. The *fire code official* shall examine or cause to be examined the accompanying *construction documents* and shall ascertain by such examinations whether the work indicated and described is in accordance with the requirements of this code.

[C] The requirements of this section are related to those found in Section 105.2.4 regarding the action of the fire code official in response to a permit application. The fire code official can delegate review of the construction documents to subordinates as provided for in Section 103.3.

[A] 106.2.1 Information on construction documents. *Construction documents* shall be drawn to scale on suitable material. Documents in a digital format are allowed to be submitted where *approved* by the *fire code official*. *Construction documents* shall be of sufficient clarity to indicate the location, nature and extent of the work proposed and show in detail that it will conform to the provisions of this code and relevant laws, ordinances, rules and regulations as determined by the *fire code official*.

[C] The construction documents are required to be of a quality and detail such that the fire code official can determine whether the work conforms to the code and other applicable laws and regulations. The permit applicant must be familiar with code requirements to prepare code-compliant construction documents. If the applicant is not familiar with the code, the construction documents will most likely not have sufficient detail to determine compliance and, thus, not be satisfactory as the basis for a permit.

[A] 106.2.2 Fire protection system shop drawings. Shop drawings for the fire protection system(s) shall be submitted to indicate compliance with this code and the *construction documents*, and shall be *approved* prior to the start of installation. Shop drawings shall contain all information as required by the referenced installation standards in Chapter 9.

[C] Since the fire protection contractor(s) may not be selected at the time a permit is issued for construction of a building, detailed shop drawings for fire protection systems are not available. Because they provide the information necessary to determine code compliance, as specified in the appropriate referenced standard in Chapter 9, they must be submitted and approved by the building official before the contractor can begin installing the system. For example, the professional responsible for the design of an automatic sprinkler system should determine that the water supply is adequate but will not be able to prepare a final set of hydraulic calculations if the specific materials and pipe sizes, lengths and arrangements have not been identified. Once the installing contractor is selected, specific hydraulic calculations can be prepared. Factors such as classification of the hazard, amount of water supply available and the density or concentration to be achieved by the system are to be included with the submission of the shop drawings. Specific data sheets identifying sprinklers, pipe dimensions, power requirements for smoke detectors, and so on, should also be included with the submission.

[A] 106.2.3 Applicant responsibility. It shall be the responsibility of the applicant to ensure that the *construction documents* include all of the fire protection requirements and the shop drawings are complete and in compliance with the applicable codes and standards.

[C] This requirement is similar to the one in Section 901.2 regarding construction documents for fire protection systems. The requirement in this section regarding shop drawings applies to all types of shop drawings, not just those for fire protection systems. The permit applicant is responsible for the review of the shop drawings, not the fire code official. The permit applicant is also responsible for seeing that the work on the job site complies with the code. Since a lot of the construction work is done in accordance with shop drawings, the applicant should review those drawings for code compliance to make sure field construction complies with the code.

[A] 106.2.4 Approved documents. *Construction documents approved* by the *fire code official* are *approved* with the intent that such *construction documents* comply in all respects with this code. Review and approval by the *fire code official* shall not relieve the applicant of the responsibility of compliance with this code.

[C] The applicant is responsible for making sure that construction complies with the code. If approved drawings include errors that do not comply with the code, the fire code official still has the authority to require that the errors be corrected. Thus, it is important that the permit applicant is familiar with code requirements to prevent preparation of construction documents that do not meet the code.

[A] 106.2.4.1 Phased approval. The *fire code official* is authorized to issue a permit for the construction of part of a structure, system or operation before the *construction documents* for the whole structure, system or operation have been submitted, provided that adequate information and detailed statements have been filed complying with pertinent requirements of this code. The holder of such permit for parts of a structure, system or operation shall proceed at the holder's own risk with the building operation and without assurance that a permit for the entire structure, system or operation will be granted.

[C] The fire code official has the authority to issue a partial permit to allow for the practice of "fast tracking" a job. Any construction under a partial permit is "at the holder's own risk" and "without assurance that a permit for the entire structure, system or operation will be granted." The fire code official is under no obligation to accept work or issue a complete permit in violation of the code, ordinances or statutes simply because a partial permit had been issued. Fast tracking puts unusual administrative and technical burdens on the fire code official. The purpose of phased approval processes is to allow construction to proceed while the design continues for other aspects of the work. Coordinating and correlating the code

aspects into the project in phases requires attention to detail and project tracking so that all code issues are addressed. The coordination of these submittals is the responsibility of the registered design professional in responsible charge described in Section 106.1.

[A] 106.3 Amended construction documents. Work shall be installed in accordance with the *approved construction documents*, and any changes made during construction that are not in compliance with the *approved construction documents* shall be resubmitted for approval as an amended set of *construction documents*.

C Any amendments to the approved construction documents must be filed before constructing the amended item. In the broadest sense, amendments include all addenda, change orders, revised drawings and marked-up shop drawings. While some minor amendments may be evaluated in the field, fire code officials should establish a policy that all amendments be submitted for review. Otherwise, a significant amendment may not be submitted because of misinterpretation, resulting in an activity that is not approved and that causes a needless delay in obtaining approval of the finished work.

[A] 106.4 Retention of construction documents. One set of *construction documents* shall be retained by the *fire code official* for a period of not less than 180 days from date of completion of the permitted work, or as required by state or local laws. One set of *approved construction documents* shall be returned to the applicant, and said set shall be kept on the site of the building or work at all times during which the work authorized thereby is in progress.

C Construction documents must be retained in case a question or dispute arises after completion of the project. Unless modified because of state or local statutes, the retention period for the approved construction documents is a minimum of 180 days following the completion of the work, typically the date the certificate of occupancy is issued. Any further retention of plans by the jurisdiction as an archival record of construction activity in the community is not required by the code. Large buildings such as commercial office buildings, hospitals and multiple-tenant mercantile facilities typically see many alterations over their lifetimes. Keeping original plans, especially if digitized, for the life of a facility will make review of future renovations much easier.

SECTION 107—TEMPORARY STRUCTURES, USES, EQUIPMENT AND SYSTEMS

[A] 107.1 General. The *fire code official* is authorized to issue a permit for temporary structures, uses, equipment or systems as required in Sections 105.5 and 105.6. Such permits shall be limited as to time of service, but shall not be permitted for more than 180 days. The *fire code official* is authorized to grant extensions for demonstrated cause.

C In the course of construction or other activities, structures that have a limited service life are often necessary. This section contains the administrative provisions that permit such temporary structures without full compliance with the code requirements for permanently occupied structures. This section should not be confused with the scope of IBC Section 3103, which regulates temporary structures larger than 120 square feet (11 m²) in area.

This section allows the fire code official to issue permits for temporary structures or uses. The applicant must specify the time period desired for the temporary structure or use, but the approval period cannot exceed 180 days. Structures or uses that are temporary but are anticipated to be in existence for more than 180 days are required to conform to code requirements for permanent structures and uses. The section also authorizes the building official to grant extensions to this time period if the applicant can provide a valid reason for the extension, which typically includes circumstances beyond the applicant's control. This provision is not intended to be used to circumvent the 180-day limitation.

[A] 107.2 Conformance. Temporary uses, equipment and systems shall conform to the requirements of this code as necessary to ensure health, safety and general welfare.

C This section makes it clear that these temporary uses, equipment and systems are to have the same level of minimum safety and protection from hazards as those that are not temporary.

[A] 107.3 Temporary service utilities. The *fire code official* is authorized to give permission to temporarily supply service utilities in accordance with Section 111.

C This section requires that Section 111 be complied with when it is necessary to connect or disconnect service utilities to execute emergency operations or eliminate an immediate hazard.

[A] 107.4 Termination of approval. The fire code official is authorized to terminate such permit for temporary uses, equipment or systems and to order the same to be discontinued.

C The fire code official can order that a temporary equipment or system be removed, or a temporary use be discontinued if conditions of the permit have been violated or the equipment, system or use poses an imminent hazard to the public, in which case the provisions of Section 115 become applicable. This text is important because it allows the fire code official to act quickly when time is of the essence in order to protect public health, safety and welfare.

SECTION 108—FEES

[A] 108.1 Fees. A permit shall not be issued until the fees have been paid, nor shall an amendment to a permit be released until the additional fee, if any, has been paid.

[C] The code anticipates that jurisdictions will establish their own fee schedules. It is intended that the fees collected by the department for permit issuance, plan review and inspection be adequate to cover the costs to the department in performing these functions. If the department has additional duties, then its budget will need to be supplemented from the general fund. This section requires that all fees be paid prior to permit issuance or release of an amendment to a permit. Since department operations are intended to be supported by fees paid by the user of department activities, it is important that these fees are received before incurring any expense. This philosophy has resulted in some departments having fees paid prior to the performance of two areas of work: plan review and inspection.

[A] 108.2 Schedule of permit fees. Where a permit is required, a fee for each permit shall be paid as required, in accordance with the schedule as established by the applicable governing authority.

[C] The jurisdiction is responsible for promulgating a schedule of fees to be charged for permits. Permit fees should be established by law, such as in an ordinance adopting the code, a separate ordinance or a legally adopted regulation, as required by state or local law. Fee schedules for operational permits should be based on an analysis of the amount of time and resources to be expended by the jurisdiction in administering the permit. This includes the amount of work to be expended in plan review, inspections and administering the permit, plus an excess to cover the department overhead.

To assist jurisdictions in establishing uniformity in fees, building evaluation data are published twice each year in the ICC's *Building Safety Journal*.

108.3 Permit valuations. The applicant for a permit shall provide an estimated value of the work for which the permit is being issued at the time of application. Such estimated valuations shall include the total value of work, including materials and labor, for which the permit is being issued, such as electrical, gas, mechanical, plumbing equipment and permanent systems. Where, in the opinion of the *fire code official*, the valuation is underestimated, the permit shall be denied unless the applicant can show detailed estimates acceptable to the *fire code official*. The *fire code official* shall have the authority to adjust the final valuation for permit fees.

[C] As indicated in Section 108.2, jurisdictions usually base their fees on the total value of the work being performed. This section requires the applicant to provide that figure, including materials and labor, for work for which the permit is sought. If the fire code official believes that the value provided by the applicant is underestimated, the permit is to be denied unless the applicant can substantiate the value by providing detailed estimates of the work to the satisfaction of the fire code official. For the construction of new buildings, the building valuation data referred above can be used by the fire code official as a yardstick against which to compare the applicant's estimate.

[A] 108.4 Work commencing before permit issuance. A person who commences any work, activity or operation regulated by this code before obtaining the necessary permits shall be subject to a fee established by the applicable governing authority, which shall be in addition to the required permit fees.

[C] The fire code official will incur certain costs (i.e., inspection time and administrative) when investigating and citing a person who has commenced work, activity or operation without having obtained a permit. The fire code official is, therefore, entitled to recover these costs by establishing a fee, in addition to that collected when the required permit is issued, to be imposed on the responsible party. Note that this is not a penalty, as described in Section 113.4; it is intended to cover the additional investigative time not included in the permit fee.

[A] 108.5 Related fees. The payment of the fee for the construction, *alteration*, removal or demolition of work done in connection to or concurrently with the work or activity authorized by a permit shall not relieve the applicant or holder of the permit from the payment of other fees that are prescribed by law.

[C] The fees for a permit may be in addition to other fees required by the jurisdiction or others for related items, such as sewer connections, water service taps, driveways and signs. It cannot be construed that the permit fee includes these other items.

[A] 108.6 Refunds. The applicable governing authority is authorized to establish a refund policy.

[C] This section allows for a refund of fees, which may be full or partial, typically resulting from the revocation, abandonment or discontinuance of a building project or operation for which a permit has been issued and fees have been collected. The refund of fees should be related to the cost of enforcement services not provided because of the termination of the project or operation. The fire code official, when authorizing a fee refund, is authorizing the disbursement of public funds; therefore, the request for a refund should be in writing and for good cause.

SECTION 109—INSPECTIONS

[A] 109.1 Inspection authority. The *fire code official* is authorized to enter and examine any building, structure, marine vessel, vehicle or premises in accordance with Section 104.4 for the purpose of enforcing this code.

C This section establishes the right of the fire code official to enter the premises in order to make the inspections required by Section 104. The right to enter structures or premises is limited for a variety of reasons.

First, to protect the right of privacy, the owner or occupant must grant the fire code official permission before an interior inspection of the property can be conducted. Permission is not required for inspections that can be accomplished from within the public right-of-way.

Second, such access may be denied by the owner or occupant. Unless the inspector has reasonable cause to believe that a code violation exists, access may be unattainable. Searches of a private residence to gather information for the purpose of enforcing codes, ordinances or regulations are considered unreasonable and are prohibited by the Fourth Amendment to the US Constitution. “Reasonable cause” in the context of this section must be distinguished from “probable cause,” which is required to gain access to property in criminal cases. The burden of proof for establishing reasonable cause may vary among jurisdictions. Usually, an inspector must show that the property is subject to inspection under the provisions of the code; that the interests of the public health, safety and welfare outweigh the individual’s right to maintain privacy; and that such an inspection is required solely to determine compliance with the provisions of the code.

Many jurisdictions do not recognize the concept of an administrative warrant and may require the fire code official to prove probable or reasonable cause in order to gain access upon refusal. This burden of proof is usually more substantial, often requiring the fire code official to stipulate in advance why access is needed (usually access is restricted to gathering evidence for seeking an indictment or making an arrest); what specific items or pieces of information are sought; its relevance to the case against the individual subject; how knowledge of the relevance of the information or items sought was obtained; and how the evidence sought will be used. In all such cases, the right to privacy must always be weighed against the right of the code official to conduct an inspection to verify that the public health, safety and welfare are not in jeopardy. Such important and complex constitutional issues should be discussed with the jurisdiction’s legal counsel. Jurisdictions should establish procedures for securing the necessary court orders when an inspection is deemed necessary following a refusal.

Third, code officials must present proper identification (see commentary, Section 105.4) and request admittance during reasonable hours—usually the normal business hours of the establishment—to be admitted.

Fourth, inspections must be aimed at securing or determining compliance with the provisions and intent of the regulations that are specifically within the established scope of the code official’s authority.

[A] 109.2 Inspections. The *fire code official* is authorized to conduct such inspections as are deemed necessary to determine the extent of compliance with the provisions of this code and to approve reports of inspection by *approved agencies* or individuals. Reports of such inspections shall be prepared and submitted in writing for review and approval. Inspection reports shall be certified by a responsible officer of such *approved agency* or by the responsible individual. The *fire code official* is authorized to engage such expert opinion as deemed necessary to report on unusual, detailed or complex technical issues subject to the approval of the governing body.

C This section establishes the fire code official’s authority to inspect buildings, structures or premises to verify that the requirements of the code are met or to accept written reports of inspections by an approved agency. The code does not, however, establish the frequency of inspections or even require that inspections be conducted, since the code does not presume to interpret or influence the adopting jurisdiction’s political, social and economic priorities. Jurisdictions may establish their inspection priorities and frequencies based on a variety of factors, including the availability of inspection resources; the level of available fire suppression services; the value of premises to the community; or the potential disruption to community services or stability if a fire occurs. In summary, each community determines and assumes its own acceptable risk level.

While a department does not have the resources to inspect every aspect of all work, the required inspections are those that are dictated by administrative rules and procedures based on many parameters, including available inspection resources. In order to expand the available resources, the fire code official may approve an inspection agency that, in his or her opinion, possesses the proper qualifications to perform the inspections. When unusual or complex technical issues arise relative to inspections, the fire code official has the authority to seek the opinion and advice of experts. A technical report from an expert requested by the building official can be used to assist in the approval process.

[A] 109.2.1 Inspection requests. It shall be the duty of the holder of the permit or their duly authorized agent to notify the *fire code official* when work is ready for inspection. It shall be the duty of the permit holder to provide access to and means for inspections of such work that are required by this code.

C It is the responsibility of the permit holder or other authorized person, such as the contractor performing the work, to arrange for the required inspections when completed work is ready and to allow for sufficient time for the fire code official

to schedule a visit to the site to prevent work from being concealed prior to being inspected. Access to the work to be inspected must be provided, including any special means such as a ladder to gain access.

[A] 109.2.2 Approval required. Work shall not be done beyond the point indicated in each successive inspection without first obtaining the approval of the *fire code official*. The *fire code official*, on notification, shall make the requested inspections and shall either indicate the portion of the construction that is satisfactory as completed, or notify the permit holder or his or her agent wherein the same fails to comply with this code. Any portions that do not comply shall be corrected, and such portion shall not be covered or concealed until authorized by the *fire code official*.

[C] This section establishes that work cannot progress beyond the point of a required inspection without approval from the fire code official. Upon making the inspection, the fire code official must either approve the completed work or notify the permit holder of portions that do not comply with the code. Approvals and notices of noncompliance must be in writing, as required by Section 113.3, to avoid any misunderstanding as to what is required. Any item not approved cannot be covered or concealed until it has been corrected and approved by the fire code official.

[A] 109.3 Concealed work. It shall be the duty of the permit applicant to cause the work to remain visible and able to be accessed for inspection purposes. Where any installation subject to inspection prior to use is covered or concealed without having first been inspected, the *fire code official* shall have the authority to require that such work be made visible and able to be accessed for inspection. Neither the *fire code official* nor the jurisdiction shall be liable for expense entailed in the removal or replacement of any material required to allow inspection.

[C] In many jurisdictions, the contractor or permit holder of an operation or a construction project is to contact the local fire inspection authority when work is completed but still exposed to allow inspection. The section requires that any work to be concealed upon completion must remain exposed until an inspection is made. If the work requiring inspection is covered up before the inspection takes place, the fire code official has the authority to require removal of the concealing construction. The section also makes it clear that any expense incurred in removing or replacing material that conceals an item to be inspected is not the responsibility of either the fire code official or the jurisdiction. Obviously, this can be a time-consuming and expensive procedure that can be eliminated by good communication and cooperation between the contractor or permit holder and the fire code official.

[A] 109.4 Approvals. Approval as the result of an inspection shall not be construed to be an approval of a violation of the provisions of this code or of other ordinances of the jurisdiction. Inspections presuming to give authority to violate or cancel provisions of this code or of other ordinances of the jurisdiction shall not be valid.

[C] As with the issuance of permits (see commentary, Section 105.3), approval as a result of an inspection is not a license to violate the code. Further, approval in violation of the code does not relieve the applicant from complying with the code and is not valid. This is significant because even if there are errors or oversights in the approval process, the permit applicant or other recipient of the inspection, not the fire code official, is still responsible for code compliance.

SECTION 110—MAINTENANCE

[A] 110.1 Maintenance of safeguards. Where any device, equipment, system, condition, arrangement, level of protection, or any other feature is required for compliance with the provisions of this code, or otherwise installed, such device, equipment, system, condition, arrangement, level of protection, or other feature shall thereafter be continuously maintained in accordance with this code and applicable referenced standards.

[C] This section does not identify who is responsible for maintenance because that determination should be made in accordance with the legal documents created between owners and occupants, such as a lease. The owner of a structure or premises, however, is usually primarily responsible for its maintenance since the owner stands to gain the most from a well-maintained property. One of the underlying assumptions is that maintaining a commercial property in good condition allows the owner to recoup a substantial portion of his or her investment in maintenance. There are three factors that may influence owners to comply with code requirements:

- Code compliance requires only a small additional investment in the property.
- The owner has a long-term interest in the property.
- The owner expects profitability after incurring the additional expense of complying with the code.

While all these factors represent economic incentives, fire code officials should be equally aware of potential disincentives to compliance, such as assessable value, expiring tax credits or historic, architectural or aesthetic criteria. The fire code official need not belabor the justifications for compliance but should be prepared to acknowledge the owner's rationalizations for failure to comply.

This section also emphasizes that any "otherwise installed" system that currently exists must be maintained. For example, an existing fire protection system cannot be removed from a building just because it is not required in new or existing buildings by current codes.

[A] 110.2 Testing and operation. Equipment requiring periodic testing or operation to ensure maintenance shall be tested or operated as specified in this code.

C Periodic testing or operation is needed to verify that the equipment can be expected to operate as needed. For example, see Section 901.6 for inspection and testing requirements for fire protection systems.

Test and inspection records must be available to the fire code official for verification that the tests and inspections required by the code and the referenced standards are in compliance.

If the fire code official requests, such records must be filed with the jurisdictional office.

[A] 110.2.1 Reinspection and testing. Where any work or installation does not pass an initial test or inspection, the necessary corrections shall be made so as to achieve compliance with this code. The work or installation shall then be resubmitted to the *fire code official* for inspection and testing.

C This section simply requires an installation to be of such quality that it will pass any tests or inspections required by the code. For example, if a fire alarm system did not pass the installation test upon completion of the system, the system is to be reworked until it passes the test.

[A] 110.3 Recordkeeping. A record of periodic inspections, tests, servicing and other operations and maintenance shall be maintained on the premises or other *approved* location for not less than 3 years, or a different period of time where specified in this code or referenced standards. Records shall be made available for inspection by the *fire code official*, and a copy of the records shall be provided to the *fire code official* on request.

The *fire code official* is authorized to prescribe the form and format of such recordkeeping. The *fire code official* is authorized to require that certain required records be filed with the *fire code official*.

C Recordkeeping of testing, inspections, servicing and other operations and maintenance are critical to the successful application of this code. Proper and consistent recordkeeping will increase the safety to both building occupants and firefighters. This section also makes it clear that records must be provided to the fire code official upon request and maintained for a period of not less than 3 years unless a different time interval is specified in the code or a referenced standard. Also, the fire code official is authorized to prescribe the form and format of such records. The recordkeeping requirements throughout the code have been specifically coordinated with this section.

[A] 110.4 Supervision. Maintenance and testing shall be under the supervision of a responsible person who shall ensure that such maintenance and testing are conducted at specified intervals in accordance with this code.

C Maintenance supervision is needed to verify that testing and general supervision is done regularly. Section 901.6 states code requirements regarding testing and maintenance of the fire protection systems.

110.5 Rendering equipment inoperable. Portable or fixed fire-extinguishing systems or devices, and fire-warning systems, shall be provided with *ready access* and shall not be rendered inoperative, except as necessary during emergencies, maintenance, repairs, *alterations*, drills or prescribed testing.

C If fire protection systems are going to be effective when needed, they must be in good operating condition. This section specifies those circumstances when they are allowed to be temporarily out of service. See Section 901.6 for code requirements regarding testing and maintenance of the fire protection systems.

[A] 110.6 Overcrowding. Overcrowding or admittance of any person beyond the *approved* capacity of a building or a portion thereof shall not be allowed. The *fire code official*, on finding any overcrowding conditions or obstructions in *aisles*, passageways or other *means of egress*, or on finding any condition that constitutes a life safety hazard, shall be authorized to cause the event to be stopped until such condition or obstruction is corrected.

C The key to enforcing this provision successfully is good judgment. Rarely is it possible to count accurately the number of people in any given place of public assembly. Usually, the proprietors or operators of such events have no accurate estimate of the crowd size or they are unwilling to provide an estimate. Despite these difficulties, when the number of people is too large to permit aisles and required egress elements to remain clear or at least flow smoothly, remedies must be sought. In many instances, overcrowding can be remedied by simply preventing any more occupants from entering, thus limiting the potential hazard to those occupants already inside. If the fire code official determines that preventing further access will be insufficient in itself, he or she is authorized to order the owner or operator to stop the event until the hazardous condition is abated, the approved occupant load is reestablished and resumption of the event is authorized by the fire code official.

SECTION 111—SERVICE UTILITIES

[A] 111.1 Authority to disconnect service utilities. The *fire code official* shall have the authority to authorize disconnection of utility service to the building, structure or system in order to safely execute emergency operations or to eliminate an immediate hazard. The *fire code official* shall notify the serving utility and, where possible, the *owner* or the *owner's* authorized agent and the occupant of the building, structure or service system of the decision to disconnect prior to taking such action. If not notified prior to disconnect-

tion, then the *owner*, the *owner's* authorized agent or occupant of the building, structure or service system shall be notified in writing as soon as practical thereafter.

C Disconnecting a service utility from the energy supply is the most radical method of hazard abatement available to the fire code official and should be reserved for cases in which all other lesser remedies have proven ineffective. Such an action must be preceded by a written notice to the owner and any occupants of the building being ordered to disconnect. Disconnection must be accomplished within the time frame established by the fire code official in the written notification. When the hazard to the public health and welfare is so imminent as to mandate immediate disconnection, the fire code official has the authority and even the obligation to cause disconnection without notice.

SECTION 112—MEANS OF APPEALS

[A] 112.1 General. In order to hear and decide appeals of orders, decisions or determinations made by the *fire code official* relative to the application and interpretation of this code, there shall be and is hereby created a board of appeals. The board of appeals shall be appointed by the applicable governing authority and shall hold office at its pleasure. The board shall adopt rules of procedure for conducting its business and shall render all decisions and findings in writing to the appellant with a duplicate copy to the *fire code official*.

C This section provides aggrieved parties with a material interest in the decisions of the fire code official a process to appeal such decisions before a board of appeals. This provides a forum, other than the court of jurisdiction, in which to review the fire code official's actions.

While this section allows literally any person to appeal a decision of the fire code official, in practice, it has been interpreted to permit appeals only by those aggrieved parties with a material or definitive interest in the decision. An aggrieved party may not appeal a code requirement per se. The intent of the appeal process is not to waive or set aside a code requirement; rather, it is intended to provide a means of reviewing a fire code official's decision on an interpretation or application of the code, or to review the equivalency of protection to the code requirements. The members of the appeals board are appointed by the "governing body" of the jurisdiction, typically a council or administrator, such as a mayor or city manager, and remain members until removed from office. The board must establish procedures for electing a chairperson, scheduling and conducting meetings, and administration. Note that Appendix A contains complete, detailed requirements for creating an appeals board, including number of members, qualifications and administrative procedures. Jurisdictions desiring to utilize these requirements must include Appendix A in their adopting ordinance.

[A] 112.2 Limitations on authority. An application for appeal shall be based on a claim that the true intent of this code or the rules legally adopted thereunder have been incorrectly interpreted, the provisions of this code do not fully apply or an equivalent or better form of construction is proposed. The board shall not have authority to waive requirements of this code.

C This section limits appeals to claims that the fire code official has misinterpreted or misapplied a code provision. The board is not allowed to set aside any of the technical requirements of the code. It is, however, allowed to consider alternative methods of compliance with the technical requirements.

[A] 112.3 Qualifications. The board of appeals shall consist of members who are qualified by experience and training on matters pertaining to the provisions of this code and are not employees of the jurisdiction.

C It is important that the decisions of the appeals board are based purely on the technical merits involved in an appeal. It is not the place for policy or political deliberations. The members of the appeals board are, therefore, expected to have experience in building construction matters. Appendix A provides more detailed qualifications for appeals board members and can be adopted by jurisdictions desiring that level of expertise.

[A] 112.4 Administration. The *fire code official* shall take action without delay in accordance with the decision of the board.

C To avoid any undue hindrance in the progress of construction, the fire code official is required to act without delay based on the board's decision. This action may be to enforce the decision or to seek judicial relief if the board's action can be demonstrated to be inappropriate.

SECTION 113—VIOLATIONS

[A] 113.1 Unlawful acts. It shall be unlawful for a person, firm or corporation to erect, construct, alter, repair, remove, demolish or utilize a building, occupancy, premises or system regulated by this code, or cause same to be done, in conflict with or in violation of any of the provisions of this code.

C Violations of the code are prohibited. This is the basis for all citations and correction notices with regard to code violations.

[A] 113.2 Owner/occupant responsibility. Correction and abatement of violations of this code shall be the responsibility of the *owner* or the *owner's* authorized agent. Where an occupant creates, or allows to be created, hazardous conditions in violation of this code, the occupant shall be held responsible for the abatement of such hazardous conditions.

C Hazards related to use and occupancy, and not those related to fixed equipment or installations, fall within the scope of the occupants' responsibility. Owners or the owner's authorized agent, however, may become liable if they allow the unlawful

operation or continuation of a public nuisance on a property under their control, especially if they knowingly or willfully lease the property in violation of fire, zoning or building regulations.

The simple rule for determining what is the responsibility of the owner or owner's authorized agents and what is the occupants' responsibility is whether the issue involves fixed equipment installations or if the structure is separate from those items related to occupancy. The owner or owner's authorized agent is usually responsible for the physical maintenance of the building or structure and its utilities and appurtenances (that is, building services and systems).

[A] 113.3 Notice of violation. Where the *fire code official* finds a building, premises, vehicle, storage facility or outdoor area that is in violation of this code, the *fire code official* is authorized to prepare a written notice of violation describing the conditions deemed unsafe and, where compliance is not immediate, specifying a time for reinspection.

C The fire code official has a duty to supply owners, agents or occupants with a written notice of code violations on the premises under their control. When possible, both the owner and the occupants should be made aware of hazardous conditions. Such notices constitute the first of several steps in the due process procedure. Violation notices must clearly indicate the defect, its location, the code section being violated, what must be done to correct the violation and the date of the reinspection. Owners, agents or occupants should also be supplied with information regarding penalties, permit applications and appeal procedures. The notice or order must be signed by the fire code official who issued it and should provide a space for the owner, agent or occupants' signature to acknowledge receipt of the document. If possible, duplicate or triplicate copies should be prepared, with the original notice issued to the responsible party. Other copies should be maintained by the inspector and the departmental record keeper.

[A] 113.3.1 Service. A notice of violation issued pursuant to this code shall be served on the *owner*, the *owner's* authorized agent, operator, occupant or other person responsible for the condition or violation, either by personal service, mail or by delivering the same to, and leaving it with, some person of responsibility on the premises. For unattended or abandoned locations, a copy of such notice of violation shall be posted on the premises in a conspicuous place at or near the entrance to such premises and the notice of violation shall be mailed by certified mail with return receipt requested or a certificate of mailing, to the last known address of the *owner*, the *owner's* authorized agent, or occupant.

C Service methods are listed by order or preference. The first and best method is to personally serve the owner, owner's authorized agent or the occupant, in that order, at the premises cited, with a signature acknowledging receipt. The next-most desirable method is service to these same parties in the order indicated at their place of business when it is not the premises cited.

While post office delivery by ordinary first-class mail is acceptable, most jurisdictions prefer certified mail with return receipt, followed by a certificate of mailing; however, owners or owner's authorized agents familiar with the legal process will often refuse to accept certified mail. As a result, many jurisdictions follow up returned certified mail with a request for a certificate of mailing. A certificate of mailing includes certification by the mail carrier or post office that the item was physically delivered to the address indicated but does not verify that the addressee actually took possession of the item. The least-desirable method of service is physically posting the premises with the violation notice. When service proves difficult, many jurisdictions pursue the mailing and posting service options simultaneously to exhaust all service methods. Jurisdictions should consult legal counsel about case law regarding legal service in their communities. The following methods are most common:

- Personal to violator.
- Personal to party at premises.
- Certified mail with return receipt.
- First-class mail with certificate of mailing.
- Posting at the premises.

[A] 113.3.2 Compliance with orders and notices. A notice of violation issued or served as provided by this code shall be complied with by the *owner*, the *owner's* authorized agent, operator, occupant or other person responsible for the condition or violation to which the notice of violation pertains.

C The party responsible for the condition that is in noncompliance is required by this section to bring the property into code compliance. See the remainder of Section 113 for what is to be done if this does not occur.

[A] 113.3.3 Prosecution of violations. If the notice of violation is not complied with promptly, the *fire code official* is authorized to request the legal counsel of the jurisdiction to institute the appropriate legal proceedings at law or in equity to restrain, correct or abate such violation or to require removal or termination of the unlawful occupancy of the structure in violation of the provisions of this code or of the order or direction made pursuant hereto.

C In situations where an unsafe condition exists and the owner or agent is not taking action to correct or eliminate the unsafe conditions, the fire code official can start legal action to remedy the situation. This action could be an eviction notice, a cease and desist order, or notification for the jurisdiction to demolish or remove the unsafe condition. As with any legal action, the attorney for the jurisdiction is to be involved to ensure that actions taken are properly handled.

[A] 113.3.4 Unauthorized tampering. Signs, tags or seals posted or affixed by the *fire code official* shall not be mutilated, destroyed or tampered with, or removed, without authorization from the *fire code official*.

[C] This section states that tampering with signs, seals or tags posted at the property is a violation of the code. The safety of occupants may depend on warning signs posted by the fire code official remaining in place.

[A] 113.4 Violation penalties. Persons who shall violate a provision of this code or shall fail to comply with any of the requirements thereof or who shall erect, install, alter, repair or do work in violation of the *approved construction documents* or directive of the *fire code official*, or of a permit or certificate used under provisions of this code, shall be guilty of a [SPECIFY OFFENSE], punishable by a fine of not more than [AMOUNT] dollars or by imprisonment not exceeding [NUMBER OF DAYS], or both such fine and imprisonment. Each day that a violation continues after due notice has been served shall be deemed a separate offense.

[C] A standard fine or other penalty as deemed appropriate by the jurisdiction is prescribed in this section. Additionally, this section identifies a principle that “each day that a violation continues shall be deemed a separate offense” for the purpose of applying the prescribed penalty in order to facilitate prompt resolution.

[A] 113.4.1 Abatement of violation. In addition to the imposition of the penalties herein described, the *fire code official* is authorized to institute appropriate action to prevent unlawful construction or to restrain, correct or abate a violation; or to prevent illegal occupancy of a structure or premises; or to stop an illegal act, conduct of business or occupancy of a structure on or about any premises.

[C] Even though the person who violated the code has paid any fine and served any sentence that may be imposed for the jurisdiction under Section 113.4, the fire code official has the right to require that the code violation be removed. If the violation is not abated, the fire code official has the right to prevent occupancy until the violation is addressed. Usually, the court will require that the violation be corrected as part of the sentence of noncompliance prior to the occupancy of the building.

SECTION 114—STOP WORK ORDER

[A] 114.1 Authority. Where the *fire code official* finds any work regulated by this code being performed in a manner contrary to the provisions of this code, or in a dangerous or unsafe manner, the *fire code official* is authorized to issue a stop work order.

[C] This section authorizes the fire code official to issue a stop work order where any work regulated by this code is being performed in a manner that is dangerous, unsafe or contrary to the provisions of this code. This section provides for the suspension of work even though a valid permit was issued, pending the removal or correction of a severe violation or unsafe condition identified by the fire code official.

Normally, correction notices, issued in accordance with Section 113.3, are used to inform the permit holder of code violations. Stop work orders are issued when enforcement can be accomplished no other way or when a dangerous condition exists.

[A] 114.2 Issuance. The stop work order shall be in writing and shall be given to the *owner* of the property, the *owner's* authorized agent or the person performing the work. Upon issuance of a stop work order, the cited work shall immediately cease. The stop work order shall state the reason for the order and the conditions under which the cited work is authorized to resume.

[C] Upon receipt of a violation notice from the fire code official, all construction activities identified in the notice must immediately cease, except as expressly permitted to correct the violation.

[A] 114.3 Emergencies. Where an emergency exists, the *fire code official* shall not be required to give a written notice prior to stopping the work.

[C] This section gives the fire code official the authority to stop the work in dispute immediately when, in his or her opinion, there is an unsafe emergency condition that has been created by the work. The need for the written notice is suspended for this situation so that the work can be stopped immediately. After the work is stopped, immediate measures should be taken to correct the work at issue.

[A] 114.4 Failure to comply. Any person who shall continue any work after having been served with a stop work order, except such work as that person is directed to perform to remove a violation or unsafe condition, shall be subject to fines established by the authority having jurisdiction.

[C] The local jurisdiction is to designate the fine that is to apply to any person who continues work that is at issue, other than abatement work. The dollar amounts for the minimum and maximum fines are to be specified in the adopting ordinance.

SECTION 115—UNSAFE STRUCTURES OR EQUIPMENT

[A] 115.1 General. If during the inspection of a premises, a structure, or any building system, in whole or in part, constitutes a clear and inimical threat to human life, safety or health, the *fire code official* shall issue such notice or orders to remove or remedy the

conditions as shall be deemed necessary in accordance with this section, and shall refer the building to the building official for any repairs, *alterations*, remodeling, removing or demolition required.

C This section describes the responsibility of the fire code official to investigate reports of unsafe structures and equipment and provides criteria for such determination.

Unsafe structures are defined as buildings or structures that are insanitary; are deficient in light, ventilation or adequate exit facilities; constitute a fire hazard; or are otherwise dangerous to human life.

This section establishes that unsafe buildings can result from illegal or improper occupancies. For example, *prima facie* evidence of an unsafe structure is an unsecured (open at door or window) vacant building. All unsafe buildings must either be demolished or made safe and secure as deemed appropriate by the fire code official.

[A] 115.1.1 Unsafe conditions. Structures or existing equipment that are or hereafter become unsafe, insanitary or deficient because of inadequate *means of egress*, inadequate light and ventilation, or that constitute a fire hazard, are otherwise dangerous to human life or the public welfare, or involve illegal or improper occupancy or inadequate maintenance, shall be deemed an unsafe condition. Unsafe structures shall be taken down and removed or made safe, as the *fire code official* deems necessary and as provided for in this section. A vacant structure that is not secured against unauthorized entry shall be deemed unsafe.

C Courts have continually upheld the right of states and their authorized subdivisions to abate public nuisances, even by demolition, and bill or assess the property owner through a tax lien for their expenses. However, care must be exercised to maintain compliance with the due process and equal protection doctrines of the Fourth and Fourteenth Amendments of the US Constitution. Jurisdictions should consult legal counsel and adopt appropriate guidelines before engaging in a nuisance abatement program.

Uninhabited or abandoned buildings that are not secured against unauthorized entry as required by Section 311 have a very high probability of intentionally set fires. When fires occur in these buildings, they present a host of unusual hazards to firefighters. Since the buildings are uninhabited, fires may develop for significant periods of time before they are detected and reported. Accordingly, such unsecured buildings are declared by this section to be unsafe and therefore subject to remediation as provided in Section 115 (see commentary, Section 311).

[A] 115.1.2 Structural hazards. Where an apparent structural hazard is caused by the faulty installation, operation or malfunction of any of the items or devices governed by this code, the *fire code official* shall immediately notify the building code official in accordance with Section 115.1.

C The fire code official is required to report structurally unsafe buildings to the building official to secure abatement of unsafe conditions. Courts have continually upheld the right of states and their authorized subdivisions to abate public nuisances, even by demolition, and bill or assess the property owner through a tax lien for their expenses. However, care must be exercised to maintain compliance with the due process and equal protection doctrines of the Fourth and Fourteenth Amendments of the US Constitution.

[A] 115.2 Evacuation. The *fire code official* or the fire department official in charge of an incident shall be authorized to order the immediate evacuation of any occupied structure deemed unsafe where such structure has hazardous conditions that pose an imminent danger to structure occupants. Persons so notified shall immediately leave the structure or premises and shall not enter or re-enter until authorized to do so by the *fire code official* or the fire department official in charge of the incident.

C The fire code official must immediately order the evacuation of any premises posing a clear and imminent threat to life or property. Structure occupants who are warned must comply with the evacuation order without delay. Upon leaving the structure, occupants may not reenter until authorization is given by the fire code official. Severe and immediate danger anticipated in this section dictates such extreme measures to protect public health, safety and welfare.

[A] 115.3 Record. The *fire code official* shall cause a report to be filed on an unsafe condition. The report shall state the occupancy of the structure and the nature of the unsafe condition.

C The fire code official must file a report on each investigation of unsafe conditions, stating the occupancy of the structure and the nature of the unsafe condition. This report provides the basis for the notice described in Section 115.4.

[A] 115.4 Notice. If an unsafe condition is found, the *fire code official* shall serve on the *owner* of the structure or the *owner's* authorized agent a written notice that describes the condition deemed unsafe and specifies the required repairs or improvements to be made to abate the unsafe condition, or requires the unsafe structure to be demolished within a stipulated time. Such notice shall require the person thus notified to declare immediately to the *fire code official* acceptance or rejection of the terms of the order.

C When a building is found to be unsafe, this information must be provided to the structure owner or agent so that they have the opportunity to fix the problem or tear down the structure. The notice should include a time frame for when the items need to be addressed. After the building owner or agent receives the report (see Section 115.3), they must inform the fire code official how they will address the issues or stipulate why they disagree with the findings.

[A] 115.5 Method of service. Such notice shall be deemed properly served where a copy thereof is served in accordance with one of the following methods:

1. A copy is delivered to the *owner* personally.

2. A copy is sent by certified or registered mail addressed to the *owner* at the last known address with return receipt requested.
3. A copy is delivered in any other manner as prescribed by local law.

If the certified or registered letter is returned showing that the letter was not delivered, a copy thereof shall be posted in a conspicuous place in or about the structure affected by such notice. Service of such notice in the foregoing manner on the *owner's* authorized agent shall constitute service of notice on the *owner*.

C The notice must be delivered to the owner in person, by certified mail or in some other lawful manner, such as delivery to a specified agent of the owner. If the owner or agent cannot be located, additional procedures are established, including posting the unsafe notice on the premises in question. Such action may be considered the equivalent of personal notice. However, it may or may not be deemed by the courts as representing a “good faith” effort to notify. In addition to complying with this section, therefore, public notice through the use of newspapers and other postings in a prominent location at the government center should be used.

[A] 115.6 Restoration or abatement. The structure or equipment determined to be unsafe by the *fire code official* is permitted to be restored to a safe condition. The *owner*, the *owner's* authorized agent, operator or occupant of a structure, premises or equipment deemed unsafe by the *fire code official* shall abate or cause to be abated or corrected such unsafe conditions either by repair, rehabilitation, demolition or other *approved* corrective action. To the extent that repairs, *alterations* or additions are made or a change of occupancy occurs during the restoration of the structure, such repairs, *alterations*, additions or change of occupancy shall comply with the requirements of Section 105.1.5 and the *International Existing Building Code*.

C This section provides that unsafe structures may be restored to a safe condition. This means that the cause of the unsafe structure notice can be abated without the structure being required to comply fully with the provisions for new construction. Any work done to eliminate the unsafe condition, as well as any change in occupancy that may occur, must comply with the IEBC.

[A] 115.7 Summary abatement. Where conditions exist that are deemed hazardous to life and property, the *fire code official* or fire department official in charge of the incident is authorized to abate summarily such hazardous conditions that are in violation of this code.

C As indicated in the commentary to Section 115.4, the fire code official is authorized to seek abatement action by the building department and bill the owner for abatement costs. Obviously, this is an extreme measure and should be done only where the owner, operator or occupant does not take such measures under the requirements of Section 115.4.

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